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DEPARTMENT OF ELECTRONICS

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BIGPROJECT REPORT

Thiết kế mạch đo nhiệt độ sử dụng Thermistor

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SUBJECT: Applied Electronics (EE3129)

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Thiết kế mạch đo nhiệt độ sử dụng Thermistor. Mạch có tầm đo $20 - 80^oC$, độ phân giải 0.1^oC, ngõ ra là điện áp tương ứng từ $0 - 5\,\mathsf{V}$:

1. Lựa chọn và mô phỏng Thermistor 2 dây (có điện trở dây dẫn là $10\,\Omega$).
2. Lựa chọn điện trở song song để tuyến tính hóa.
3. Lựa chọn OPAMP sử dụng, mô phỏng OPAMP theo datasheet.
4. Đảm bảo sai số (giữa giá trị lý thuyết và thực tế đo qua mạch) $\pm 5\%$.
5. Thiết kế mạch đọc điện áp cho ngõ ra ADC. Tiến hành lựa chọn ADC (số bit, nguồn). Mô phỏng mạch đọc và kiểm chứng.

Chapter 1 Thực hiện mạch đọc điện áp cho ngõ ra ADC

1.1 Tổng quan về ADC

Theo chuẩn ngõ ra điện áp đọc từ Sensor là từ $0 \rightarrow 5\,\mathsf{V}$, với ngõ vào là từ $20^oC \rightarrow 80^oC$ với bước nhảy là 0.1^oC. Thực hiện tính giá trị của một bước nhảy nhỏ nhất là:

$$\frac{5-0}{\frac{80-20}{0.1}+1} \approx 8.3195\,\mathsf{mV}$$

Vậy có nghĩa là $\Delta_{min} \approx 8.3195\,\mathsf{mV}$, từ đó ta có:

$$2^Q = \frac{V_{ref^+} - V_{ref^-}}{\Delta_{min}}$$

Giả sử, chọn $V_{ref^+} = 5\,\mathsf{V}$ và $V_{ref^-} = 0\,\mathsf{V}$. Ta có $Q \approx 9.23\,\mathsf{bit}$

Tuy nhiên, trong thị trường không có ADC $9.23\,\mathsf{bit}$ cho nên nhóm em sẽ chọn con có trên thị trường là ADS1115IDGSR.

1.2 Electrical Characteristics của ADS1115

- Ultra-small packages:
  + X2QFN: $2\,\mathsf{mm} \times 1.5\,\mathsf{mm} \times 0.4\,\mathsf{mm}$
  + SOT: $2.9\,\mathsf{mm} \times 2.8\,\mathsf{mm} \times 0.6\,\mathsf{mm}$
- Wide supply range: $2.0\,\mathsf{V}$ to $5.5\,\mathsf{V}$
- Low current consumption: $150\,\mu\mathsf{A}$ (continuous-conversion mode)
- Programmable data rate: 8SPS to 860SPS
- Single-cycle settling
- Internal low-drift voltage reference
- Internal oscillator
- I2C interface: four pin-selectable addresses

- Operating temperature range: -40^oC to $+125^oC$
- Family of devices:
  + ADS1115: four single-ended or two differential inputs with comparator and PGA

Từ những thông tin, ADC ADS1115 có tầm hoạt động ở dải nhiệt độ phù hợp trong khoảng $20^oC \rightarrow 80^oC$, nhóm emm sử dụng nguồn cấp là $V_{DD} = 5.5\,\mathsf{V}$ và cho phép nguồn sai số trong khoảng từ $[5.3, 5.5]\,\mathsf{V}$ phù hợp với thông số dưới đây:

DIGITAL INPUT/OUTPUT						
V_{IH}	High-level input voltage		0.7 VDD		5.5	V
V_{IL}	Low-level input voltage		GND		0.3 VDD	V
V_{OL}	Low-level output voltage	I_{OL} = 3mA	GND	0.15	0.4	V
	Input leakage current	GND < V_{DIG} < VDD	–10		10	µA

Hình 1.1: Digital Input/Output của ADS1115.

Như vậy, với $GND \approx 0\,\mathsf{V} < V_{IN} < V_{DD} = 5.5\,\mathsf{V}$ thì có thể phù hợp cho tầm đo từ $0 \rightarrow 5\,\mathsf{V}$ từ mạch đọc giá trị của sensor.

Tiếp theo, ta cần quan tâm đến giá trị như sau:

SYSTEM PERFORMANCE						
	Resolution (no missing codes)			16		Bits
DR	Data rate			8, 16, 32, 64, 128, 250, 475, 860		SPS
	Data rate variation	All data rates		–10%	10%	
	Output noise			See *Noise Performance* section		
INL	Integral nonlinearity	DR = 8SPS, FSR = ±2.048V(2)			1	LSB
	Offset error	FSR = ±2.048V, differential inputs	3	±1	3	LSB
		FSR = ±2.048V, single-ended inputs		±3		
	Offset drift over temperature	FSR = ±2.048V		0.005		LSB/°C
	Long-term Offset drift	FSR = ±2.048V, T_A = 125°C, 1000 hours		±1		LSB
	Offset power-supply rejection	FSR = ±2.048V, DC supply variation		1		LSB/V
	Offset channel match	Match between any two inputs		3		LSB
	Gain error(3)	FSR = ±2.048V, T_A = 25°C		0.01%	0.15%	
	Gain drift over temperature(3)	FSR = ±0.256V		7		ppm/°C
		FSR = ±2.048V		5	40	
		FSR = ±6.144V(1)		5		
	Long-term gain drift(3)	FSR = ±2.048V, T_A = 125°C, 1000 hours		±0.05%		
	Gain power-supply rejection			80		ppm/V
	Gain match(3)	Match between any two gains		0.02%	0.1%	
	Gain channel match	Match between any two inputs		0.05%	0.1%	
CMRR	Common-mode rejection ratio	At DC, FSR = ±0.256V		105		dB
		At DC, FSR = ±2.048V		100		
		At DC, FSR = ±6.144V(1)		90		
		f_{CM} = 60Hz, DR = 8SPS		105		
		f_{CM} = 50Hz, DR = 8SPS		105		

Hình 1.2: Electrical Characteristics của ADS1115.

- Data rate (DR) là giá trị dữ liệu Digital được đổi theo đơn vị thời gian. Với $\mathsf{DR} = 128$ Sample per second là giá trị default của ADS1115, với thời gian cách nhau đọc dữ liệu khoảng tầm $8\,\mathsf{ms}$.

- Integral Nonlinearity (INL) là thước đo độ lệch của đặc tuyến chuyển đổi thực tế so với đặc tuyến lý tưởng trong các bộ chuyển đổi ADC/DAC, sau khi đã loại bỏ sai số offset và gain. Với $\mathsf{INL} = 1\,\mathsf{LSB}$.

- Offset error là sai số khi sử dụng chế độ Single-Ended với giá trị là $\pm 3\,\mathsf{LSB}$.

- Gain drift over temperature là sai số của phép đo khi có ảnh hưởng của nhiệt độ với tại $\mathsf{FSR} = \pm 6.144\,\mathsf{V}$ thì Gain drift $= 5ppm/^oC$ tức là mỗi ^oC sẽ gây sai số là 5 phần ngàn.

- Common-mode Rejection Ratio (CMRR) là thước đo khả năng của một bộ khuếch đại vi sai trong việc loại bỏ tín hiệu chung xuất hiện đồng thời và cùng pha trên cả hai đầu vào. Với $\mathsf{CMRR} = 90\,dB$.

1.3 Typical Characteristics của ADS1115

- Single-Ended Offset Error vs Temperature:

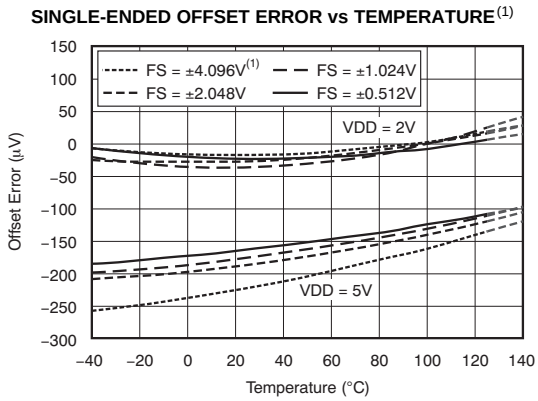

Figure 4.

+ Tại $T = 20^oC$: Offset Error $= -225\,\mu\text{V}$

+ Tại $T = 80^oC$: Offset Error $= -120\,\mu\text{V}$

- Gain Error vs Temperature:

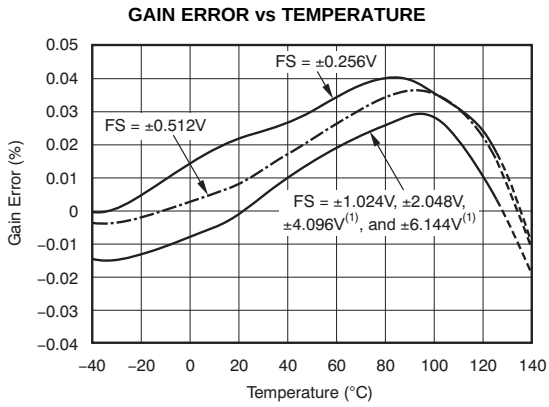

Figure 6.

+ Tại $T = 20^oC$: Gain Error $= 0.00\,\%$

+ Tại $T = 80^oC$: Gain Error $= 0.03\,\%$

- Gain Error vs Supply:

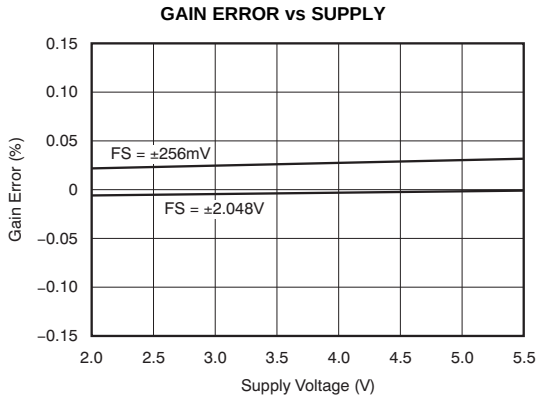

Figure 7.

+ Tại $V_{DD} = 5.5\,\text{V}$: Gain Error $= 0.00\,\%$

- INL vs Supply Voltage:

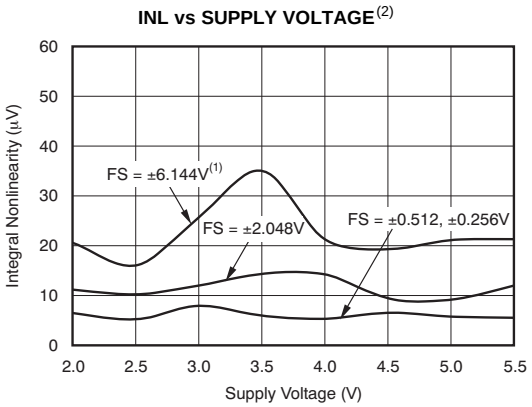

Figure 8.

+ Tại $V_{DD} = 5.5\,\mathsf{V}$: INL vs Voltage = $20\,\mu\mathsf{V}$

- INL vs Temperature:

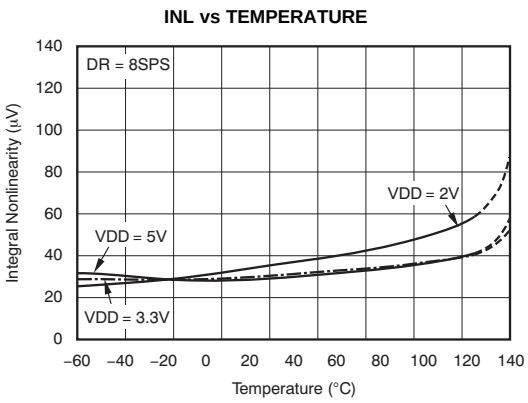

Figure 13.

+ Tại $T = 20^oC$: INL vs Temperature = $22\,\mu\mathsf{V}$

+ Tại $T = 80^oC$: INL vs Temperature = $38\,\mu\mathsf{V}$

- Data Rate vs Temperature:

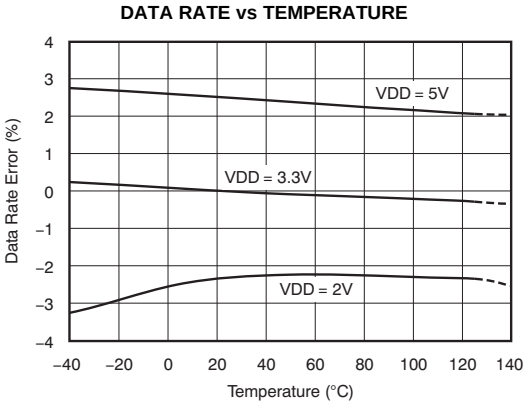

Figure 20.

+ Tại $T = 20^oC$: Data Rate Error = $2.8\,\%$

+ Tại $T = 80^oC$: Data Rate Error = $1.2\,\%$

1.4 Data Format

Với ADS1115 là một ADC 16-bit với ngõ ra là một chuỗi bit có dấu theo giá trị như sau:

Table 4. Input Signal versus Ideal Output Code

INPUT SIGNAL, V_{IN} (AIN_P – AIN_N)	IDEAL OUTPUT CODE(1)
≥ FS $(2^{15} – 1)/2^{15}$	7FFFh
$+FS/2^{15}$	0001h
0	0
$–FS/2^{15}$	FFFFh
≤ –FS	8000h

1. Excludes the effects of noise, INL, offset, and gain errors.

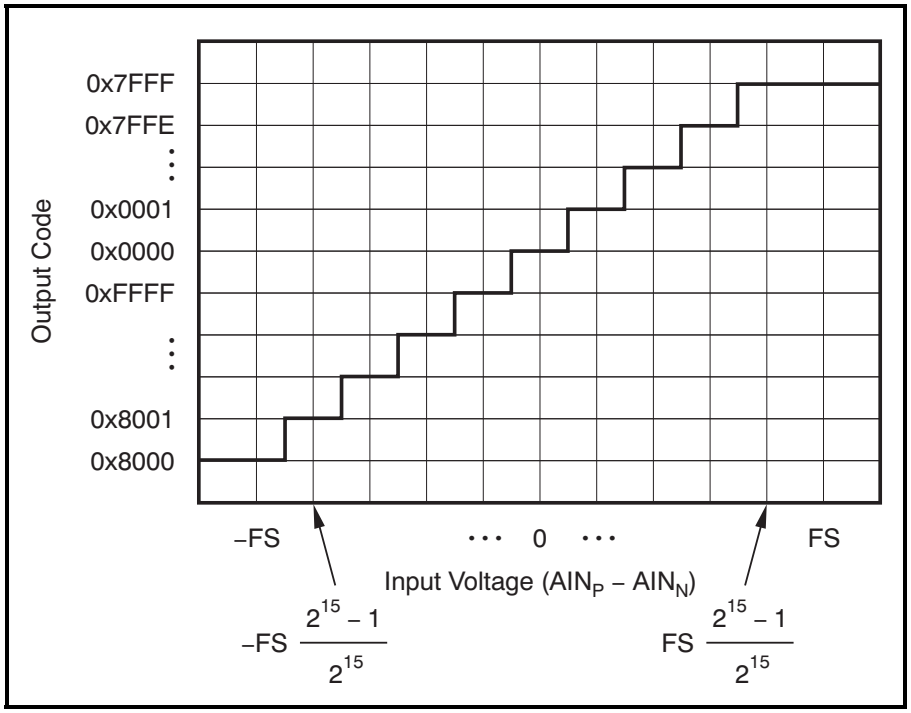

Figure 27. ADS1113/4/5 Code Transition Diagram

Hình 1.3: ADS1115 Code Transition Diagram.

Theo hình 1.3, với FS = 6.144 như ở trên thì giá trị lớn nhất mà ADC có thể đọc được

là $V_{ref} \times \dfrac{2^{15}-1}{2^{15}} \approx 6.1438\,\mathsf{V}$ vẫn phù hợp với tầm đo là từ $0 \to 5\,\mathsf{V}$. Từ đó ta có công thức suy ngược từ D_{out} nhận được từ ADC là:

$$V_{out} = \frac{V_{ref}}{2^{15}} \times D_{out} = \frac{6.1438}{32768} \times D_{out}$$

1.5 Kết quả của việc tính toán ADC

Ta có bảng tính toán giá bên dưới, trong bảng giá trị có các giá trị như sau:

- **T_s :** là giá trị nhiệt độ chuẩn cần đo.
- **V_i :** là giá trị điện áp đo được từ sensor trong khoảng từ $0 \to 5\,\mathsf{V}$ với sai số là 5%.
- **Offset_Error :** là giá trị sai số của ADS1115 do nhiệt độ tạo nên.
- **Gain_Error :** là giá trị của độ lợi của ADS1115 do nhiệt độ tạo nên.
- **V_i_ADC :** là giá trị mà ADC thực sự nhận tại chân `IN0`.
- **D_out :** là giá trị tính toán từ giá trị điện đọc từ chân `IN0` với công thức là $\dfrac{V_i_ADC \times 2^{15}}{FS}$.
- **D_cal_Error :** là giá trị mà thực sự ADS1115 cho ra ở ngõ ra với các giá trị sai số cộng thêm do chính ADC và nhiệt độ.
- **Cal_V :** là giá trị được tính trong MCU chuyển đổi từ giá trị Digital của ADS1115 gửi về cho MCU.
- **Cal_Temp :** là giá trị chuyển đổi từ điện áp tính toán được `Cal_V` về giá trị nhiệt độ theo công thức chuẩn là $T = \dfrac{V_{in} + \frac{5}{3}}{\frac{1}{2}}$.

Requirement				
ERROR	5%			

ADC				
	Min	Typ	Max	Unit
VDD	5.2	5.5	5.5	V
GND	-0.3	0	0.3	V
Resolution		16		Bit
Analog_In_V_to_GND				V
Temperature	-60		150	độ C
PGA (program gain amplifier)	0.666666667	0.666866667		
FS (Full-Scale)	-6.142157353	6.1438125	6.142157353	V
Common-Mode Input Impedance		6000000		Ohm
DR (Data Rate)		128		SPS
Range_Temperature	20		80	độ C
Offset Voltage	-0.000225		-0.00012	V
Gain Error	0%		0.03%	%
Gain Drift		0.000500%		%/độ C
	Tại FS = 6.144	0.030000%		
Common Mode Rejection	Tại FS = 6.144	90		dB
INL_vs_VoltageSupply		0.000020		V
INL_vs_Temperature	0.000022		0.000038	V
Digital Input/Output				
V_IH	3.85		5.5	
V_IL	-0.5		1.65	

Calibration				
	Value_Standard	Value_Measurement	Error	
raw_zero_25	0	0	0	
raw_full_25	5	5	0	
raw_zero_20	0	0	0	
raw_full_20	5	5	0	
raw_zero_50	0	0	0	
raw_full_50	5	5	0	
raw_zero_80	0	0	0	
raw_full_80	5	5	0	

Digtal output				
Input Signal (V_i)	>= FS(2^15-1)/(2^15)	6.141969909	V	
Offset Error	-3		3	LSB
Offset Drift		0.005		LSB/độ C
	FS = 6.144V	0.3		LSB
Offset power supply rejection		1		LSB/V
	VDD = 5.5V	0.3		LSB
INL		1		LSB

T_s	V_i	Offset_Vol	Gain_Error	G_E	V_i_ADC	Error_V_i_ADC	D_out	D_cal_error	BIN	Cal_V	Cal_Temp	Error_re		Error_ab	
20	0.017	-0.000225	0.000000%	-0.001500	0.017257201	1.5129%	92.0688	94	0000000001011110	0.017624462	20.2114935	1%	3.6988%	0.211494	1.839692
20.1	0.019954682	-0.000224	0.000100%	-0.001469	0.020207083	1.2649%	107.8067	111	0000000001101111	0.020811865	20.2497424	1%	0.0055%	0.149742	0.002645
20.2	0.025642402	-0.000224	0.000100%	-0.001439	0.025887233	0.9548%	138.1109	137	0000000010001001	0.025686716	20.3082406	1%		0.108241	
20.3	0.032330742	-0.000223	0.000200%	-0.001408	0.032565971	0.7276%	173.7426	175	0000000010101111	0.032811499	20.393738	0%		0.093738	
20.4	0.038074328	-0.000223	0.000200%	-0.001378	0.038302627	0.5996%	204.3482	202	0000000011001010	0.037873844	20.4544861	0%		0.054486	
20.5	0.044816929	-0.000222	0.000300%	-0.001347	0.045036343	0.4896%	240.2732	240	0000000011110000	0.044998627	20.5399835	0%		0.039984	
20.6	0.05158625	-0.000222	0.000300%	-0.001317	0.051798108	0.4107%	276.3479	280	0000000100011000	0.052498398	20.6299808	0%		0.029981	
20.7	0.058382283	-0.000221	0.000400%	-0.001286	0.058586016	0.3490%	312.562	313	0000000100111001	0.058685709	20.7042285	0%		0.004229	
20.8	0.06520502	-0.000220	0.000400%	-0.001256	0.065400949	0.3005%	348.9204	347	0000000101011011	0.065060514	20.7807262	0%		0.019274	
20.9	0.072054452	-0.000220	0.000500%	-0.001225	0.072244024	0.2631%	385.4288	386	0000000110000010	0.072372791	20.8684735	0%		0.031527	
21	0.078930568	-0.000219	0.000500%	-0.001195	0.079113097	0.2313%	422.076	423	0000000110100111	0.07931008	20.951721	0%		0.048279	
21.1	0.084867512	-0.000219	0.000600%	-0.001164	0.085045567	0.2098%	453.7264	454	0000000111000110	0.085122402	21.0214688	0%		0.078531	
21.2	0.091797781	-0.000218	0.000600%	-0.001134	0.091969553	0.1871%	490.6664	489	0000000111101001	0.091684702	21.1002164	0%		0.099784	
21.3	0.098754705	-0.000218	0.000700%	-0.001103	0.098921658	0.1691%	527.7566	532	0000000100010100	0.099746956	21.1969635	0%		0.103037	
21.4	0.105738272	-0.000217	0.000700%	-0.001073	0.105899704	0.1527%	564.9851	570	0000001000111010	0.106871738	21.2824609	1%		0.117539	
21.5	0.112748471	-0.000216	0.000800%	-0.001042	0.112904885	0.1387%	602.3584	605	0000001001011101	0.113434038	21.3612085	1%		0.138792	
21.6	0.118823527	-0.000216	0.000800%	-0.001012	0.118977181	0.1293%	634.7547	639	0000001001111111	0.119808844	21.4377061	1%		0.162294	
21.7	0.125887774	-0.000215	0.000900%	-0.000981	0.126037188	0.1187%	672.4205	672	0000001010100000	0.125996155	21.5119539	1%		0.188046	
21.8	0.132978615	-0.000215	0.000900%	-0.000951	0.133125068	0.1101%	710.235	709	0000001011000101	0.132933443	21.5952013	1%		0.204799	
21.9	0.140096038	-0.000214	0.001000%	-0.000920	0.140239071	0.1021%	748.1889	749	0000001011101101	0.140433214	21.6851986	1%		0.214801	
22	0.146281565	-0.000214	0.001000%	-0.000890	0.1464233	0.0969%	781.1824	785	0000001100010001	0.147183008	21.7661961	1%		0.233804	
22.1	0.153452935	-0.000213	0.001100%	-0.000859	0.153592049	0.0907%	819.4283	819	0000001100110011	0.153557814	21.8426938	1%		0.257306	
22.2	0.160650843	-0.000212	0.001100%	-0.000829	0.160787598	0.0851%	857.8173	857	0000001101011001	0.160682596	21.9281912	1%		0.271809	
22.3	0.166919306	-0.000212	0.001200%	-0.000798	0.167056042	0.0819%	891.26	890	0000001101111010	0.166869907	22.0024389	1%		0.297561	
22.4	0.174171077	-0.000211	0.001200%	-0.000768	0.174306255	0.0776%	929.9406	934	0000001110100110	0.175119656	22.1014359	1%		0.298564	
22.5	0.181449339	-0.000211	0.001300%	-0.000737	0.181584554	0.0745%	968.7711	970	0000001111001010	0.18186945	22.1824334	1%		0.317567	
22.6	0.187800619	-0.000210	0.001300%	-0.000707	0.187935789	0.0720%	1002.6555	1004	0000001111101100	0.188244255	22.2589311	2%		0.341069	
22.7	0.195132651	-0.000209	0.001400%	-0.000676	0.195267689	0.0692%	1041.7719	1040	0000010000010000	0.194994049	22.3399286	2%		0.360071	
22.8	0.202491122	-0.000209	0.001400%	-0.000646	0.202627262	0.0672%	1081.0359	1083	0000010000111011	0.203056303	22.4366756	2%		0.363324	
22.9	0.208925084	-0.000208	0.001500%	-0.000615	0.209062546	0.0658%	1115.3688	1115	0000010001011011	0.20905612	22.5086734	2%		0.391327	
23	0.216337223	-0.000208	0.001500%	-0.000585	0.216476618	0.0644%	1154.9236	1154	0000010010000010	0.216368397	22.5964208	2%		0.403579	
23.1	0.222826509	-0.000207	0.001600%	-0.000554	0.222968017	0.0635%	1189.5558	1189	0000010010100101	0.222930696	22.6751684	2%		0.424832	
23.2	0.230292244	-0.000207	0.001600%	-0.000524	0.230436526	0.0627%	1229.401	1228	0000010011001100	0.230242973	22.7629157	2%		0.437084	
23.3	0.236836788	-0.000206	0.001700%	-0.000493	0.236983983	0.0622%	1264.3323	1265	0000010011110001	0.237180262	22.8461631	2%		0.453837	
23.4	0.244356043	-0.000205	0.001700%	-0.000463	0.244505864	0.0613%	1304.4623	1308	0000010100011100	0.245242516	22.9429102	2%		0.457090	
23.5	0.251901602	-0.000205	0.001800%	-0.000432	0.252055739	0.0612%	1344.7416	1347	0000010101000011	0.252554792	23.0306575	2%		0.469342	
23.6	0.258528469	-0.000204	0.001800%	-0.000402	0.2586865	0.0611%	1380.1173	1379	0000010101100011	0.258554609	23.1026553	2%		0.497345	
23.7	0.266127425	-0.000204	0.001900%	-0.000371	0.266290652	0.0613%	1420.6862	1424	0000010110010000	0.266991852	23.2039022	2%		0.496098	
23.8	0.272809362	-0.000203	0.001900%	-0.000341	0.272977297	0.0616%	1456.3601	1460	0000010110110100	0.273741646	23.2848997	2%		0.515100	
23.9	0.28046163	-0.000203	0.002000%	-0.000310	0.280635651	0.0620%	1497.2182	1497	0000010111011001	0.280678934	23.3681472	2%		0.531853	
24	0.287198558	-0.000202	0.002000%	-0.000280	0.287378108	0.0625%	1533.1898	1533	0000010111111101	0.287428728	23.4491447	2%		0.550855	
24.1	0.293963383	-0.000201	0.002100%	-0.000249	0.294149154	0.0632%	1569.314	1574	0000011000100110	0.295115993	23.5413919	2%		0.558608	
24.2	0.300756084	-0.000201	0.002100%	-0.000219	0.300949189	0.0642%	1605.5929	1609	0000011001001001	0.301678293	23.6201395	2%		0.579860	
24.3	0.307576642	-0.000200	0.002200%	-0.000188	0.307776791	0.0651%	1642.0188	1644	0000011001101100	0.308240593	23.6988871	2%		0.601113	
24.4	0.315363091	-0.000200	0.002200%	-0.000158	0.31557124	0.0660%	1683.6029	1688	0000011010011000	0.316490341	23.7978841	2%		0.602116	
24.5	0.322238425	-0.000199	0.002300%	-0.000127	0.32245448	0.0670%	1720.3257	1721	0000011010111001	0.322677652	23.8721318	3%		0.627868	
24.6	0.329141548	-0.000199	0.002300%	-0.000097	0.329366605	0.0684%	1757.2025	1760	0000011011100000	0.329989929	23.9598792	3%		0.640121	
24.7	0.337007873	-0.000198	0.002400%	-0.000066	0.337241619	0.0694%	1799.2165	1802	0000011100001010	0.337864689	24.0543763	3%		0.645624	
24.8	0.34396563	-0.000197	0.002400%	-0.000036	0.344208241	0.0705%	1836.3841	1838	0000011100101110	0.344614483	24.1353738	3%		0.664626	
24.9	0.350951106	-0.000197	0.002500%	-0.000005	0.351204351	0.0722%	1873.709	1877	0000011101010101	0.35192676	24.2231211	3%		0.676879	
25	0.358897075	-0.000196	0.002500%	0.000025	0.359160052	0.0733%	1916.1534	1921	0000011110000001	0.360176508	24.3221181	3%		0.677882	
25.1	0.365937033	-0.000196	0.002600%	0.000056	0.366211538	0.0750%	1953.7738	1954	0000011110100010	0.366363819	24.3963658	3%		0.703634	
25.2	0.373004635	-0.000195	0.002600%	0.000086	0.373289733	0.0764%	1991.5366	1993	0000011111001001	0.373676096	24.4841132	3%		0.715887	
25.3	0.381030001	-0.000195	0.002700%	0.000117	0.381327609	0.0781%	2034.4195	2035	0000011111110011	0.381550856	24.5786103	3%		0.721390	
25.4	0.388151924	-0.000194	0.002700%	0.000147	0.388461019	0.0796%	2072.4769	2071	0000100000010111	0.38830065	24.6596078	3%		0.740392	
25.5	0.395301412	-0.000193	0.002800%	0.000178	0.395622822	0.0813%	2110.6858	2115	0000100001000011	0.396550398	24.7586048	3%		0.741395	
25.6	0.402478435	-0.000193	0.002800%	0.000208	0.402813208	0.0832%	2149.0472	2148	0000100001100100	0.402737709	24.8328525	3%		0.767147	
25.7	0.410609553	-0.000192	0.002900%	0.000239	0.410957758	0.0848%	2192.4992	2191	0000100010001111	0.410799963	24.9295996	3%		0.770400	
25.8	0.417840671	-0.000192	0.002900%	0.000269	0.418203152	0.0868%	2231.154	2230	0000100010110110	0.41811224	25.0173469	3%		0.782653	
25.9	0.42509924	-0.000191	0.003000%	0.000300	0.425475865	0.0886%	2269.9546	2275	0000100011100011	0.426549482	25.1185938	3%		0.781406	
26	0.432385229	-0.000191	0.003000%	0.000330	0.432777026	0.0906%	2308.907	2310	0000100100000110	0.433111782	25.1973414	3%		0.802659	
26.1	0.43969861	-0.000190	0.003100%	0.000361	0.440105467	0.0925%	2348.005	2348	0000100100101100	0.440236565	25.2828388	3%		0.817161	
26.2	0.447961449	-0.000189	0.003100%	0.000391	0.448383744	0.0943%	2392.1704	2395	0000100101011011	0.449048796	25.3885855	3%		0.811414	
26.3	0.455328618	-0.000189	0.003200%	0.000422	0.455767927	0.0965%	2431.5657	2434	0000100110000010	0.456361073	25.4763329	3%		0.823667	
26.4	0.462723084	-0.000188	0.003200%	0.000452	0.463178415	0.0984%	2471.1014	2474	0000100110101010	0.463860844	25.5663301	3%		0.833670	
26.5	0.470144816	-0.000188	0.003300%	0.000483	0.470618097	0.1007%	2510.7928	2512	0000100111010000	0.470985626	25.6518275	3%		0.848172	
26.6	0.477593781	-0.000187	0.003300%	0.000513	0.478084009	0.1026%	2550.6242	2551	0000100111110111	0.478297903	25.7395748	3%		0.860425	
26.7	0.485069947	-0.000186	0.003400%	0.000544	0.48557807	0.1048%	2590.6057	2589	0000101000011101	0.485422686	25.8250722	3%		0.874928	
26.8	0.49257328	-0.000186	0.003400%	0.000574	0.493100286	0.1070%	2630.7374	2629	0000101001000101	0.492922457	25.9150695	3%		0.884931	
26.9	0.500103746	-0.000185	0.003500%	0.000605	0.500649604	0.1091%	2671.0138	2674	0000101001110010	0.501359699	26.0163164	3%		0.883684	
27	0.507661311	-0.000185	0.003500%	0.000635	0.508226998	0.1114%	2711.4399	2714	0000101010011010	0.50885947	26.1063136	3%		0.893686	
27.1	0.51524594	-0.000184	0.003600%	0.000666	0.515831445	0.1136%	2752.0104	2754	0000101011000010	0.516359241	26.1963109	3%		0.903689	
27.2	0.522857598	-0.000184	0.003600%	0.000696	0.523463889	0.1160%	2792.7302	2796	0000101011101100	0.524234001	26.290808	3%		0.909192	
27.3	0.530496248	-0.000183	0.003700%	0.000727	0.531123333	0.1182%	2833.594	2838	0000101100010110	0.532108761	26.3853051	3%		0.914695	
27.4	0.537250699	-0.000182	0.003700%	0.000757	0.537897843	0.1205%	2869.7367	2872	0000101100111000	0.538483566	26.4618028	3%		0.938197	
27.5	0.544944147	-0.000182	0.003800%	0.000788	0.545614045	0.1229%	2910.9034	2915	0000101101100011	0.54654582	26.5585498	3%		0.941450	
27.6	0.552664479	-0.000181	0.003800%	0.000818	0.553356077	0.1251%	2952.2079	2955	0000101111000011	0.554045591	26.6485471	3%		0.951453	
27.7	0.560411656	-0.000181	0.003900%	0.000849	0.561127004	0.1276%	2993.6665	2994	0000101111010010	0.561357868	26.7362944	3%		0.963706	
27.8	0.56818564	-0.000180	0.003900%	0.000879	0.568923673	0.1299%	3035.2625	3035	0000101111011011	0.569045134	26.8285416	3%		0.971458	
27.9	0.575986392	-0.000180	0.004000%	0.000910	0.576749181	0.1324%	3077.0123	3079	0000110000000111	0.577294882	26.9275386	3%		0.972461	
28	0.583813873	-0.000179	0.004000%	0.000940	0.584600343	0.1347%	3118.899	3119	0000110000101111	0.584794653	27.0175358	4%		0.982464	

28.1	0.591668042	-0.000178	0.004100%	0.000971	0.592479283	0.1371%	3160.9339	3163	0000110001011011	0.593044401	27.1165328	3%		0.983467	
28.2	0.598645126	-0.000178	0.004100%	0.001001	0.599481148	0.1397%	3198.2896	3199	0000110001111111	0.599794195	27.1975303	4%		1.002470	
28.3	0.606553484	-0.000177	0.004200%	0.001032	0.607415277	0.1421%	3240.6189	3240	0000110010101000	0.607481461	27.2897775	4%		1.010222	
28.4	0.614488409	-0.000177	0.004200%	0.001062	0.615376878	0.1446%	3283.0948	3287	0000110011010111	0.616293692	27.3955243	4%		1.004476	
28.5	0.622449859	-0.000176	0.004300%	0.001093	0.623365134	0.1470%	3325.7129	3327	0000110011111111	0.623793463	27.4855216	4%		1.014478	
28.6	0.630437792	-0.000176	0.004300%	0.001123	0.631380767	0.1496%	3368.4771	3373	0000110100101101	0.6324182	27.5890184	4%		1.010982	
28.7	0.637553119	-0.000175	0.004400%	0.001154	0.638522908	0.1521%	3406.5811	3408	0000110101010000	0.638980499	27.667766	4%		1.032234	
28.8	0.645594828	-0.000174	0.004400%	0.001184	0.646592325	0.1545%	3449.6322	3449	0000110101111001	0.646667765	27.7600132	4%		1.039987	
28.9	0.65366289	-0.000174	0.004500%	0.001215	0.654690268	0.1572%	3492.8355	3494	0000110110100110	0.655105007	27.8612601	4%		1.038740	
29	0.661757259	-0.000173	0.004500%	0.001245	0.66281339	0.1596%	3536.1732	3538	0000110111010010	0.663354755	27.9602571	4%		1.039743	
29.1	0.668982628	-0.000173	0.004600%	0.001276	0.670068561	0.1623%	3574.8802	3575	0000110111110111	0.670292044	28.0435045	4%		1.056495	
29.2	0.677130426	-0.000172	0.004600%	0.001306	0.67824614	0.1648%	3618.5084	3622	0000111000100110	0.679104275	28.1492513	4%		1.050749	
29.3	0.685304397	-0.000172	0.004700%	0.001337	0.686452106	0.1675%	3662.288	3661	0000111001001101	0.686416552	28.2369986	4%		1.063001	
29.4	0.69261208	-0.000171	0.004700%	0.001367	0.693789411	0.1700%	3701.4332	3702	0000111001110110	0.694103817	28.3292458	4%		1.070754	
29.5	0.700839208	-0.000170	0.004800%	0.001398	0.702048591	0.1726%	3745.4967	3748	0000111010100100	0.702728554	28.4327426	4%		1.067257	
29.6	0.709092369	-0.000170	0.004800%	0.001428	0.710334644	0.1752%	3789.7036	3789	0000111011001101	0.710415819	28.5249898	4%		1.075010	
29.7	0.716481964	-0.000169	0.004900%	0.001459	0.717756086	0.1778%	3829.2977	3829	0000111011110101	0.71791559	28.6149871	4%		1.085013	
29.8	0.724788002	-0.000169	0.004900%	0.001489	0.726096073	0.1805%	3873.7924	3873	0000111100100001	0.726165339	28.7139841	4%		1.086016	
29.9	0.732232296	-0.000168	0.005000%	0.001520	0.733573239	0.1831%	3913.6838	3914	0000111101001010	0.733852604	28.8062312	4%		1.093769	
30	0.740591019	-0.000168	0.005000%	0.001550	0.741966978	0.1858%	3958.4652	3963	0000111101111011	0.743039824	28.9164779	4%		1.083522	
30.1	0.748089823	-0.000167	0.005100%	0.001581	0.74949969	0.1885%	3998.6529	4000	0000111110100000	0.749977112	28.9997253	4%		1.100275	
30.2	0.755616293	-0.000166	0.005100%	0.001611	0.757059823	0.1910%	4038.9869	4039	0000111111000111	0.757289389	29.0874727	4%		1.112527	
30.3	0.76405416	-0.000166	0.005200%	0.001642	0.765535073	0.1938%	4084.2032	4085	0000111111110101	0.765914125	29.1909695	4%		1.109030	
30.4	0.771634854	-0.000165	0.005200%	0.001672	0.773150464	0.1964%	4124.8321	4123	0001000000011011	0.773038908	29.2764669	4%		1.123533	
30.5	0.779243069	-0.000165	0.005300%	0.001703	0.780795666	0.1992%	4165.6199	4166	0001000001000110	0.781101162	29.3732139	4%		1.126786	
30.6	0.787759631	-0.000164	0.005300%	0.001733	0.789349473	0.2018%	4211.2553	4210	0001000001110010	0.78935091	29.4722109	4%		1.127789	
30.7	0.795421774	-0.000163	0.005400%	0.001764	0.797048665	0.2045%	4252.3313	4251	0001000010011011	0.797038176	29.5644581	4%		1.135542	
30.8	0.803990219	-0.000163	0.005400%	0.001794	0.805656463	0.2072%	4298.2547	4302	0001000011001110	0.806600384	29.6792046	4%		1.120795	
30.9	0.811706085	-0.000162	0.005500%	0.001825	0.813410453	0.2100%	4339.623	4338	0001000011110010	0.813350178	29.7602021	4%		1.139798	
31	0.820326202	-0.000162	0.005500%	0.001855	0.822071036	0.2127%	4385.828	4388	0001000100100100	0.822724892	29.8726987	4%		1.127301	
31.1	0.828095583	-0.000161	0.005600%	0.001886	0.829879626	0.2154%	4427.4876	4432	0001000101010000	0.83097464	29.9716957	4%		1.128304	
31.2	0.835892131	-0.000161	0.005600%	0.001916	0.837716084	0.2182%	4469.2959	4468	0001000101110100	0.837724434	30.0526932	4%		1.147307	
31.3	0.84458984	-0.000160	0.005700%	0.001947	0.846455777	0.2209%	4515.923	4517	0001000110100101	0.846911654	30.1629398	4%		1.137060	
31.4	0.852439586	-0.000159	0.005700%	0.001977	0.854345512	0.2236%	4558.0154	4559	0001000111001111	0.854786413	30.257437	4%		1.142563	
31.5	0.860316339	-0.000159	0.005800%	0.002008	0.86226465	0.2265%	4600.2648	4599	0001000111110111	0.862286184	30.3474342	4%		1.152566	
31.6	0.869962248	-0.000158	0.005800%	0.002038	0.871955175	0.2291%	4651.9647	4657	0001001000110001	0.873160852	30.4779302	4%		1.122070	
31.7	0.877890886	-0.000158	0.005900%	0.002069	0.879927336	0.2320%	4694.4969	4697	0001001001011001	0.880660624	30.5679275	4%		1.132073	
31.8	0.885846367	-0.000157	0.005900%	0.002099	0.887925	0.2346%	4737.1652	4741	0001001010000101	0.888910372	30.6669245	4%		1.133076	
31.9	0.893828637	-0.000157	0.006000%	0.002130	0.895951891	0.2375%	4779.9895	4783	0001001010101111	0.896785131	30.7614216	4%		1.138578	
32	0.902704792	-0.000156	0.006000%	0.002160	0.904873193	0.2402%	4827.5855	4828	0001001011011100	0.905222374	30.8626685	4%		1.137332	
32.1	0.910739479	-0.000155	0.006100%	0.002191	0.912952631	0.2430%	4870.6901	4873	0001001100001001	0.913659616	30.9639154	4%		1.136085	
32.2	0.918800785	-0.000155	0.006100%	0.002221	0.921059329	0.2458%	4913.9401	4915	0001001100110011	0.921534376	31.0584125	4%		1.141587	
32.3	0.926888652	-0.000154	0.006200%	0.002252	0.929193064	0.2486%	4957.3343	4958	0001001101011110	0.92959663	31.1551596	4%		1.144840	
32.4	0.935003022	-0.000154	0.006200%	0.002282	0.937353931	0.2514%	5000.8733	5002	0001001110001010	0.937846378	31.2541565	4%		1.145843	
32.5	0.944006025	-0.000153	0.006300%	0.002313	0.946405927	0.2542%	5049.1666	5050	0001001110111010	0.946846104	31.3621532	4%		1.137847	
32.6	0.952172229	-0.000153	0.006300%	0.002343	0.954619768	0.2570%	5092.9882	5095	0001001111100111	0.955283346	31.4634002	3%		1.136600	
32.7	0.959504569	-0.000152	0.006400%	0.002374	0.961998215	0.2599%	5132.353	5134	0001010000001110	0.962595623	31.5511475	4%		1.148853	
32.8	0.967724366	-0.000151	0.006400%	0.002404	0.970265746	0.2626%	5176.461	5179	0001010000111011	0.971032866	31.6523944	3%		1.147606	
32.9	0.975970371	-0.000151	0.006500%	0.002435	0.978562028	0.2655%	5220.7225	5223	0001010001100111	0.979282614	31.7513914	3%		1.148609	
33	0.984242523	-0.000150	0.006500%	0.002465	0.986883047	0.2683%	5265.1159	5269	0001010010010101	0.987907351	31.8548882	3%		1.145112	
33.1	0.992540761	-0.000150	0.006600%	0.002496	0.995232717	0.2712%	5309.6622	5308	0001010010111100	0.995219627	31.9426355	3%		1.157364	
33.2	1.000865022	-0.000149	0.006600%	0.002526	1.003606987	0.2740%	5354.3398	5358	0001010011101110	1.004594341	32.0551321	3%		1.144868	
33.3	1.009215245	-0.000149	0.006700%	0.002557	1.012009806	0.2769%	5399.1696	5401	0001010100011001	1.012656595	32.1518791	3%		1.148121	
33.4	1.017591368	-0.000148	0.006700%	0.002587	1.020437089	0.2797%	5444.13	5445	0001010101000101	1.020906343	32.2508761	3%		1.149124	
33.5	1.025993327	-0.000147	0.006800%	0.002618	1.028891814	0.2825%	5489.2368	5489	0001010101110001	1.029156092	32.3498731	3%		1.150127	
33.6	1.034421058	-0.000147	0.006800%	0.002648	1.037372868	0.2854%	5534.484	5535	0001010110011111	1.037780828	32.4533699	3%		1.146630	
33.7	1.042874498	-0.000146	0.006900%	0.002679	1.045880256	0.2882%	5579.8717	5584	0001010111010000	1.046968048	32.5636166	3%		1.136383	
33.8	1.051353582	-0.000146	0.006900%	0.002709	1.054413832	0.2911%	5625.3992	5625	0001010111111001	1.054655313	32.6558638	3%		1.144136	
33.9	1.059858245	-0.000145	0.007000%	0.002740	1.062973634	0.2939%	5671.0665	5673	0001011000101001	1.063655039	32.7638605	3%		1.136140	
34	1.067541328	-0.000145	0.007000%	0.002770	1.070710035	0.2968%	5712.341	5717	0001011001010101	1.071904787	32.8628574	3%		1.137143	
34.1	1.076097968	-0.000144	0.007100%	0.002801	1.079322989	0.2997%	5758.2919	5761	0001011010000001	1.080154535	32.9618544	3%		1.138146	
34.2	1.084679989	-0.000143	0.007100%	0.002831	1.087960842	0.3025%	5804.3757	5803	0001011010101011	1.088029295	33.0563515	3%		1.143648	
34.3	1.093287325	-0.000143	0.007200%	0.002862	1.096626704	0.3054%	5850.6089	5854	0001011011011110	1.097591503	33.171098	3%		1.128902	
34.4	1.101919908	-0.000142	0.007200%	0.002892	1.105316315	0.3082%	5896.9688	5900	0001011100001100	1.10621624	33.2745949	3%		1.125405	
34.5	1.110577672	-0.000142	0.007300%	0.002923	1.114033823	0.3112%	5943.4776	5944	0001011100111000	1.114465988	33.3735919	3%		1.126408	
34.6	1.118419569	-0.000141	0.007300%	0.002953	1.121931462	0.3140%	5985.6122	5985	0001011101100001	1.122153254	33.465839	3%		1.134161	
34.7	1.127128509	-0.000140	0.007400%	0.002984	1.130700346	0.3169%	6032.3951	6034	0001011110010010	1.131340473	33.5760857	3%		1.123914	
34.8	1.135862423	-0.000140	0.007400%	0.003014	1.139494685	0.3198%	6079.3137	6083	0001011111000011	1.140527693	33.6863323	3%		1.113668	
34.9	1.143783339	-0.000139	0.007500%	0.003045	1.147474221	0.3227%	6121.8853	6126	0001011111101110	1.148589947	33.7830794	3%		1.116921	
35	1.152568018	-0.000139	0.007500%	0.003075	1.156320525	0.3256%	6169.0812	6168	0001100000011000	1.156464706	33.8775765	3%		1.122424	
35.1	1.160541614	-0.000138	0.007600%	0.003106	1.164353917	0.3285%	6211.9401	6217	0001100001001001	1.165651926	33.9878231	3%		1.112177	
35.2	1.168541964	-0.000138	0.007600%	0.003136	1.172414473	0.3314%	6254.9439	6258	0001100001110010	1.173339191	34.0800703	3%		1.119930	
35.3	1.177402781	-0.000137	0.007700%	0.003167	1.181338898	0.3343%	6302.5566	6302	0001100010011110	1.18158894	34.1790673	3%		1.120933	
35.4	1.186288151	-0.000136	0.007700%	0.003197	1.190287314	0.3371%	6350.2973	6353	0001100011010001	1.191151148	34.2938138	3%		1.106186	
35.5	1.194366278	-0.000136	0.007800%	0.003228	1.198428618	0.3401%	6393.7319	6395	0001100011111011	1.199025908	34.3883109	3%		1.111689	
35.6	1.203301566	-0.000135	0.007800%	0.003258	1.207428178	0.3429%	6441.7454	6442	0001100100101010	1.207838139	34.4940577	3%		1.105942	
35.7	1.211431535	-0.000135	0.007900%	0.003289	1.215622527	0.3460%	6485.4631	6487	0001100101010111	1.216275381	34.5953046	3%		1.104695	
35.8	1.220416452	-0.000134	0.007900%	0.003319	1.224672949	0.3488%	6533.7479	6537	0001100110001001	1.225650095	34.7078011	3%		1.092199	
35.9	1.228597976	-0.000134	0.008000%	0.003350	1.232920064	0.3518%	6577.7471	6578	0001100110110010	1.23333736	34.8000483	3%		1.099952	
36	1.237632229	-0.000133	0.008000%	0.003380	1.242021064	0.3546%	6626.3018	6631	0001100111100111	1.243274557	34.9192947	3%		1.080705	
36.1	1.245865017	-0.000132	0.008100%	0.003411	1.250319659	0.3576%	6670.5756	6672	0001101000010000	1.250961823	35.0115419	3%		1.088458	
36.2	1.254948308	-0.000132	0.008100%	0.003441	1.259471952	0.3605%	6719.404	6722	0001101001000010	1.260336536	35.1240384	3%		1.075962	

36.3	1.263232066	-0.000131	0.008200%	0.003472	1.267822745	0.3634%	6763.9562	6768	0001101001110000	1.268961273	35.2275353	3%		1.072465	
36.4	1.272364096	-0.000131	0.008200%	0.003502	1.277025034	0.3663%	6813.0513	6817	0001101010100001	1.278148493	35.3377819	3%		1.062218	
36.5	1.280698525	-0.000130	0.008300%	0.003533	1.285427736	0.3693%	6857.8806	6856	0001101011001000	1.28546077	35.4255292	3%		1.074471	
36.6	1.289058699	-0.000130	0.008300%	0.003563	1.293856503	0.3722%	6902.8488	6901	0001101011110101	1.293898012	35.5267761	3%		1.073224	
36.7	1.298263789	-0.000129	0.008400%	0.003594	1.303134044	0.3751%	6952.3454	6952	0001101100101000	1.30346022	35.6415226	3%		1.058477	
36.8	1.306674182	-0.000128	0.008400%	0.003624	1.311613258	0.3780%	6997.5828	6998	0001101101010110	1.312084957	35.7450195	3%		1.054981	
36.9	1.315110094	-0.000128	0.008500%	0.003655	1.320120924	0.3810%	7042.9721	7042	0001101110000010	1.320334705	35.8440165	3%		1.055984	
37	1.324387549	-0.000127	0.008500%	0.003685	1.329471437	0.3839%	7092.8579	7092	0001101110110100	1.329709419	35.956513	3%		1.043487	
37.1	1.332873218	-0.000127	0.008600%	0.003716	1.338030123	0.3869%	7138.5194	7140	0001101111100100	1.338709145	36.0645097	3%		1.035490	
37.2	1.340570175	-0.000126	0.008600%	0.003746	1.345795307	0.3898%	7179.9474	7182	0001110000001110	1.346583904	36.1590069	3%		1.040993	
37.3	1.349107395	-0.000126	0.008700%	0.003777	1.35440677	0.3928%	7225.8903	7229	0001110000111101	1.355396135	36.2647536	3%		1.035246	
37.4	1.358481641	-0.000125	0.008700%	0.003807	1.363856623	0.3957%	7276.3062	7278	0001110001101110	1.364583355	36.3750003	3%		1.025000	
37.5	1.367067994	-0.000124	0.008800%	0.003838	1.372517493	0.3986%	7322.5128	7324	0001110010011110	1.373208092	36.4784971	3%		1.021503	
37.6	1.375679321	-0.000124	0.008800%	0.003868	1.381203592	0.4016%	7368.854	7368	0001110011001000	1.38145784	36.5774941	3%		1.022506	
37.7	1.384315544	-0.000123	0.008900%	0.003899	1.389915599	0.4045%	7415.3334	7416	0001110011111000	1.390457565	36.6854908	3%		1.014509	
37.8	1.393784257	-0.000123	0.008900%	0.003929	1.399463522	0.4075%	7466.2724	7470	0001110100101110	1.400582256	36.8069871	3%		1.993013	
37.9	1.402468973	-0.000122	0.009000%	0.003960	1.408225316	0.4104%	7513.0175	7513	0001110101011001	1.40864451	36.9037341	3%		0.996266	
38	1.411178344	-0.000122	0.009000%	0.003990	1.41701199	0.4134%	7559.8952	7562	0001110110001010	1.41783173	37.0139808	3%		0.986019	
38.1	1.419912286	-0.000121	0.009100%	0.004021	1.425824293	0.4164%	7606.9097	7608	0001110110111000	1.426456467	37.1174776	3%		0.982522	
38.2	1.429474163	-0.000120	0.009100%	0.004051	1.435467005	0.4192%	7658.3545	7662	0001110111101110	1.436581158	37.2389739	3%		0.961026	
38.3	1.438255946	-0.000120	0.009200%	0.004082	1.444329462	0.4223%	7705.6365	7707	0001111000011011	1.4450184	37.3402208	3%		0.959779	
38.4	1.447062055	-0.000119	0.009200%	0.004112	1.453214435	0.4252%	7753.0387	7754	0001111001001010	1.453830631	37.4459676	2%		0.954032	
38.5	1.455892409	-0.000119	0.009300%	0.004143	1.46212676	0.4282%	7800.5868	7805	0001111001111101	1.463392839	37.5607141	2%		0.939286	
38.6	1.464746923	-0.000118	0.009300%	0.004173	1.47106142	0.4311%	7848.2541	7853	0001111010101101	1.472392565	37.6687108	2%		0.931289	
38.7	1.472827387	-0.000117	0.009400%	0.004204	1.479220788	0.4341%	7891.7851	7891	0001111011010011	1.479517347	37.7542082	2%		0.945792	
38.8	1.481731037	-0.000117	0.009400%	0.004234	1.488206859	0.4370%	7939.7267	7938	0001111100000010	1.488329578	37.8599549	2%		0.940045	
38.9	1.490658597	-0.000116	0.009500%	0.004265	1.497217984	0.4400%	7987.8019	7991	0001111100110111	1.498266775	37.9792013	2%		0.920799	
39	1.499609982	-0.000116	0.009500%	0.004295	1.50625309	0.4430%	8036.0051	8036	0001111101100100	1.506704018	38.0804482	2%		0.919552	
39.1	1.508585107	-0.000115	0.009600%	0.004326	1.515313101	0.4460%	8084.3411	8089	0001111110011001	1.516641214	38.1996946	2%		0.900305	
39.2	1.516791096	-0.000115	0.009600%	0.004356	1.523600648	0.4489%	8128.5559	8131	0001111111000011	1.524515974	38.2941917	2%		0.905808	
39.3	1.525814514	-0.000114	0.009700%	0.004387	1.532710259	0.4519%	8177.1566	8179	0001111111110011	1.533515699	38.4021884	2%		0.897812	
39.4	1.534070765	-0.000113	0.009700%	0.004417	1.54104832	0.4548%	8221.641	8223	0010000000011111	1.541765448	38.5011854	2%		0.898815	
39.5	1.543142133	-0.000113	0.009800%	0.004448	1.550208201	0.4579%	8270.5098	8271	0010000001001111	1.550765173	38.6091821	2%		0.890818	
39.6	1.55223681	-0.000112	0.009800%	0.004478	1.559389496	0.4608%	8319.4929	8320	0010000010000000	1.559952393	38.7194287	2%		0.880571	
39.7	1.56135471	-0.000112	0.009900%	0.004509	1.568597253	0.4639%	8368.6172	8367	0010000010101111	1.568764624	38.8251755	2%		0.874825	
39.8	1.569709386	-0.000111	0.009900%	0.004539	1.577036291	0.4668%	8413.6402	8417	0010000011100001	1.578139338	38.9376721	2%		0.862328	
39.9	1.57887454	-0.000111	0.010000%	0.004570	1.586292633	0.4698%	8463.0237	8468	0010000100010100	1.587701546	39.0524185	2%		0.847581	
40	1.588062653	-0.000110	0.010000%	0.004600	1.595570009	0.4727%	8512.5194	8516	0010000101000100	1.596701271	39.1604153	2%		0.839585	
40.1	1.596490504	-0.000109	0.010100%	0.004631	1.604085757	0.4757%	8557.9517	8557	0010000101101101	1.604388536	39.2526624	2%		0.847338	
40.2	1.605725342	-0.000109	0.010100%	0.004661	1.613412186	0.4787%	8607.7091	8606	0010000110011110	1.613575756	39.3629091	2%		0.837091	
40.3	1.614982872	-0.000108	0.010200%	0.004692	1.622762602	0.4817%	8657.5945	8656	0010000111010000	1.62295047	39.4754056	2%		0.824594	
40.4	1.623483103	-0.000108	0.010200%	0.004722	1.631352073	0.4847%	8703.4202	8708	0010001000000100	1.632700172	39.5924021	2%		0.807598	
40.5	1.632786822	-0.000107	0.010300%	0.004753	1.640750031	0.4877%	8753.5592	8754	0010001000110010	1.641324909	39.6958989	2%		0.804101	
40.6	1.641335221	-0.000107	0.010300%	0.004783	1.649388971	0.4907%	8799.6487	8798	0010001001011110	1.649574657	39.7948959	2%		0.805104	
40.7	1.650684768	-0.000106	0.010400%	0.004814	1.658834117	0.4937%	8850.0395	8851	0010001010010011	1.659511854	39.9141423	2%		0.785858	
40.8	1.659280977	-0.000105	0.010400%	0.004844	1.667521168	0.4966%	8896.3858	8901	0010001011000101	1.668886568	40.0266388	2%		0.773361	
40.9	1.668675989	-0.000105	0.010500%	0.004875	1.677014151	0.4997%	8947.0317	8947	0010001011110011	1.677511305	40.1301357	2%		0.769864	
41	1.677319644	-0.000104	0.010500%	0.004905	1.685749963	0.5026%	8993.6381	8996	0010001100100100	1.686698524	40.2403823	2%		0.759618	
41.1	1.686759755	-0.000104	0.010600%	0.004936	1.695289418	0.5057%	9044.532	9043	0010001101010011	1.695510756	40.3461291	2%		0.753871	
41.2	1.694679247	-0.000103	0.010600%	0.004966	1.70329854	0.5086%	9087.2615	9087	0010001101111111	1.703760504	40.445126	2%		0.754874	
41.3	1.703395014	-0.000103	0.010700%	0.004997	1.712111143	0.5117%	9134.2776	9139	0010001110110011	1.713510206	40.5621225	2%		0.737878	
41.4	1.712903706	-0.000102	0.010700%	0.005027	1.721718493	0.5146%	9185.5337	9190	0010001111100110	1.723072414	40.676869	2%		0.723131	
41.5	1.721666002	-0.000101	0.010800%	0.005058	1.730577972	0.5176%	9232.7999	9234	0010010000010010	1.731322163	40.775866	2%		0.724134	
41.6	1.730451967	-0.000101	0.010800%	0.005088	1.73946105	0.5206%	9280.1919	9279	0010010000111111	1.739759405	40.8771129	2%		0.722887	
41.7	1.740027319	-0.000100	0.010900%	0.005119	1.749138864	0.5236%	9331.824	9333	0010010001110101	1.749884096	40.9986092	2%		0.701391	
41.8	1.748859255	-0.000100	0.010900%	0.005149	1.758069255	0.5266%	9379.4685	9382	0010010010100110	1.759071316	41.1088558	2%		0.691144	
41.9	1.758478215	-0.000099	0.011000%	0.005180	1.767792077	0.5297%	9431.3407	9432	0010010011011000	1.76844603	41.2213524	2%		0.678648	
42	1.767355744	-0.000099	0.011000%	0.005210	1.77676941	0.5326%	9479.2356	9479	0010010100000111	1.777258261	41.3270991	2%		0.672901	
42.1	1.776256472	-0.000098	0.011100%	0.005241	1.785771396	0.5357%	9527.2621	9526	0010010100110110	1.786070492	41.4328459	2%		0.667154	
42.2	1.785940674	-0.000097	0.011100%	0.005271	1.795559764	0.5386%	9579.484	9584	0010010101110000	1.79694516	41.5633419	2%		0.636658	
42.3	1.794886427	-0.000097	0.011200%	0.005302	1.804609157	0.5417%	9627.7634	9632	0010010110100000	1.805944885	41.6713386	1%		0.628661	
42.4	1.803855094	-0.000096	0.011200%	0.005332	1.813679323	0.5446%	9676.1536	9675	0010010111001011	1.814007139	41.7680857	1%		0.631914	
42.5	1.812846581	-0.000096	0.011300%	0.005363	1.822775817	0.5477%	9724.6842	9728	0010011000000000	1.823944336	41.887332	1%		0.612668	
42.6	1.821860791	-0.000095	0.011300%	0.005393	1.831892876	0.5507%	9773.3246	9773	0010011000101101	1.832381578	41.9885789	1%		0.611421	
42.7	1.830897626	-0.000094	0.011400%	0.005424	1.841035086	0.5537%	9822.0992	9821	0010011001011101	1.841381304	42.0965756	1%		0.603424	
42.8	1.839956993	-0.000094	0.011400%	0.005454	1.850199665	0.5567%	9870.9931	9871	0010011010001111	1.850756018	42.2090722	1%		0.590928	
42.9	1.849038792	-0.000093	0.011500%	0.005485	1.859388218	0.5597%	9920.0149	9923	0010011011000011	1.86050572	42.3260686	1%		0.573931	
43	1.857391304	-0.000093	0.011500%	0.005515	1.867843139	0.5627%	9965.1227	9965	0010011011101101	1.86838048	42.4205658	1%		0.579434	
43.1	1.866518773	-0.000092	0.011600%	0.005546	1.87707873	0.5658%	10014.3955	10017	0010011100100001	1.878130182	42.5375622	1%		0.562438	
43.2	1.875668383	-0.000092	0.011600%	0.005576	1.886336271	0.5688%	10063.7854	10068	0010011101010100	1.88769239	42.6523087	1%		0.547691	
43.3	1.884840035	-0.000091	0.011700%	0.005607	1.895617437	0.5718%	10113.3013	10112	0010011110000000	1.895942139	42.7513057	1%		0.548694	
43.4	1.894033632	-0.000090	0.011700%	0.005637	1.904919335	0.5747%	10162.9278	10162	0010011110110010	1.905316853	42.8638022	1%		0.536198	
43.5	1.903249074	-0.000090	0.011800%	0.005668	1.914246696	0.5778%	10212.6902	10216	0010011111110100	1.915441544	42.9852985	1%		0.514701	
43.6	1.911741221	-0.000089	0.011800%	0.005698	1.922844257	0.5808%	10258.559	10263	0010100000010111	1.924253775	43.0910453	1%		0.508955	
43.7	1.921001154	-0.000089	0.011900%	0.005729	1.932217496	0.5839%	10308.5661	10312	0010100001001000	1.933440994	43.2012919	1%		0.498708	
43.8	1.930282635	-0.000088	0.011900%	0.005759	1.941610033	0.5868%	10358.6762	10359	0010100001110111	1.942253225	43.3070387	1%		0.492961	
43.9	1.939585564	-0.000088	0.012000%	0.005790	1.951027678	0.5899%	10408.9202	10411	0010100010101011	1.952002928	43.4240351	1%		0.475965	
44	1.948169194	-0.000087	0.012000%	0.005820	1.959719423	0.5929%	10455.2915	10455	0010100011010111	1.960252676	43.5230321	1%		0.476968	
44.1	1.957515816	-0.000086	0.012100%	0.005851	1.969181156	0.5959%	10505.7708	10507	0010100100001011	1.970002378	43.6400285	1%		0.459971	
44.2	1.966883585	-0.000086	0.012100%	0.005881	1.978663766	0.5989%	10556.3614	10555	0010100100111011	1.979002104	43.7480252	1%		0.451975	
44.3	1.975535055	-0.000085	0.012200%	0.005912	1.987427383	0.6020%	10603.1162	10603	0010100101101011	1.988001829	43.8560219	1%		0.443978	
44.4	1.984945914	-0.000085	0.012200%	0.005942	1.996954472	0.6050%	10653.9441	10654	0010100110011110	1.997564037	43.9707684	1%		0.429232	

44.5	1.994377615	-0.000084	0.012300%	0.005973	2.006504117	0.6080%	10704.8924	10708	0010100111010100	2.007688728	44.0922647	1%		0.407735	
44.6	2.003096018	-0.000084	0.012300%	0.006003	2.01533573	0.6110%	10752.0099	10756	0010101000000100	2.016688454	44.2002614	1%		0.399739	
44.7	2.011837264	-0.000083	0.012400%	0.006034	2.024191888	0.6141%	10799.2583	10800	0010101000110000	2.024938202	44.2992584	1%		0.400742	
44.8	2.020601253	-0.000082	0.012400%	0.006064	2.033069435	0.6171%	10846.6209	10847	0010101001011111	2.033750433	44.4050052	1%		0.394995	
44.9	2.030118609	-0.000082	0.012500%	0.006095	2.042708564	0.6202%	10898.0466	10902	0010101010010110	2.044062618	44.5287514	1%		0.371249	
45	2.038926672	-0.000081	0.012500%	0.006125	2.05163156	0.6231%	10945.6516	10945	0010101011000001	2.052124872	44.6254985	1%		0.374502	
45.1	2.049214211	-0.000081	0.012600%	0.006156	2.062046814	0.6262%	11001.218	11005	0010101011111101	2.063374529	44.7604943	1%		0.339506	
45.2	2.058064832	-0.000080	0.012600%	0.006186	2.071013765	0.6292%	11049.0576	11054	0010101100101110	2.072561749	44.870741	1%		0.329259	
45.3	2.067663996	-0.000080	0.012700%	0.006217	2.080737581	0.6323%	11100.9351	11105	0010101101100001	2.082123957	44.9854875	1%		0.314513	
45.4	2.076557867	-0.000079	0.012700%	0.006247	2.089749168	0.6352%	11149.0127	11152	0010101110010000	2.090936188	45.0912343	1%		0.308766	
45.5	2.085473762	-0.000078	0.012800%	0.006278	2.098785573	0.6383%	11197.2228	11202	0010101111000010	2.100310902	45.2037308	1%		0.296269	
45.6	2.095134571	-0.000078	0.012800%	0.006308	2.108571072	0.6413%	11249.4294	11248	0010101111110000	2.108935638	45.3072277	1%		0.292772	
45.7	2.104093093	-0.000077	0.012900%	0.006339	2.117651519	0.6444%	11297.8745	11296	0010110000100000	2.117935364	45.4152244	1%		0.284776	
45.8	2.113073326	-0.000077	0.012900%	0.006369	2.126753248	0.6474%	11346.4331	11350	0010110001010110	2.128060055	45.5367207	1%		0.263279	
45.9	2.122794843	-0.000076	0.013000%	0.006400	2.136602729	0.6505%	11398.981	11403	0010110010001011	2.137997252	45.655967	1%		0.244033	
46	2.131817076	-0.000076	0.013000%	0.006430	2.145747862	0.6535%	11447.7712	11448	0010110010111000	2.146434494	45.7572139	1%		0.242786	
46.1	2.140860705	-0.000075	0.013100%	0.006461	2.154916245	0.6565%	11496.6854	11495	0010110011100111	2.155246725	45.8629607	1%		0.237039	
46.2	2.149925626	-0.000074	0.013100%	0.006491	2.164104452	0.6595%	11545.7053	11548	0010110100011100	2.165183922	45.9822071	0%		0.217793	
46.3	2.159011732	-0.000074	0.013200%	0.006522	2.173317732	0.6626%	11594.8591	11599	0010110101001111	2.17474613	46.0969536	0%		0.203046	
46.4	2.167404779	-0.000073	0.013200%	0.006552	2.181830754	0.6656%	11640.2769	11639	0010110101110111	2.182245901	46.1869508	0%		0.213049	
46.5	2.177247078	-0.000073	0.013300%	0.006583	2.191806358	0.6687%	11693.4977	11697	0010110110110001	2.193120569	46.3174468	0%		0.182553	
46.6	2.186396106	-0.000072	0.013300%	0.006613	2.201081474	0.6717%	11742.9813	11743	0010110111011111	2.201745306	46.4209437	0%		0.179056	
46.7	2.195565895	-0.000071	0.013400%	0.006644	2.210380277	0.6747%	11792.5913	11797	0010111000010101	2.211869997	46.54244	0%		0.157560	
46.8	2.20475634	-0.000071	0.013400%	0.006674	2.219699225	0.6778%	11842.3088	11847	0010111001000111	2.221244711	46.6549365	0%		0.145063	
46.9	2.213967332	-0.000070	0.013500%	0.006705	2.22904066	0.6808%	11892.1462	11897	0010111001111001	2.230619425	46.7674331	0%		0.132567	
47	2.223198765	-0.000070	0.013500%	0.006735	2.238402012	0.6838%	11942.0899	11946	0010111010101010	2.239806644	46.8776797	0%		0.122320	
47.1	2.232450531	-0.000069	0.013600%	0.006766	2.247785657	0.6869%	11992.1526	11997	0010111011011101	2.249368853	46.9924262	0%		0.107574	
47.2	2.241722524	-0.000069	0.013600%	0.006796	2.257188986	0.6899%	12042.3203	12047	0010111100001111	2.258743567	47.1049228	0%		0.095077	
47.3	2.250310476	-0.000068	0.013700%	0.006827	2.265905416	0.6930%	12088.8233	12088	0010111100111000	2.266430832	47.19717	0%		0.102830	
47.4	2.259623707	-0.000067	0.013700%	0.006857	2.275350387	0.6960%	12139.2131	12139	0010111101101011	2.27599304	47.3119165	0%		0.088084	
47.5	2.268956841	-0.000067	0.013800%	0.006888	2.284819277	0.6991%	12189.7305	12190	0010111110011110	2.285555248	47.426663	0%		0.073337	
47.6	2.278309769	-0.000066	0.013800%	0.006918	2.294305373	0.7021%	12240.3398	12244	0010111111010100	2.295679939	47.5481593	0%		0.051841	
47.7	2.288382104	-0.000066	0.013900%	0.006949	2.30451981	0.7052%	12294.8348	12295	0011000000000111	2.305242147	47.6629058	0%		0.037094	
47.8	2.297074574	-0.000065	0.013900%	0.006979	2.313341983	0.7082%	12341.9019	12341	0011000000110101	2.313866884	47.7664026	0%		0.033597	
47.9	2.306486233	-0.000065	0.014000%	0.007010	2.322892301	0.7113%	12392.8538	12392	0011000001101000	2.323429092	47.8811491	0%		0.018851	
48	2.315917251	-0.000064	0.014000%	0.007040	2.332459357	0.7143%	12443.8949	12446	0011000010011110	2.333553783	48.0026454	0%		0.002645	
48.1	2.32467224	-0.000063	0.014100%	0.007071	2.341348505	0.7174%	12491.3194	12494	0011000011001110	2.342553509	48.1106421	0%		0.010642	
48.2	2.33344859	-0.000063	0.014100%	0.007101	2.350258363	0.7204%	12538.8543	12538	0011000011111010	2.350803257	48.2096391	0%		0.009639	
48.3	2.34293925	-0.000062	0.014200%	0.007132	2.359889568	0.7235%	12590.2377	12595	0011000100110011	2.361490431	48.3378852	0%		0.037885	
48.4	2.351756885	-0.000062	0.014200%	0.007162	2.368842116	0.7265%	12638.0004	12641	0011000101100001	2.370115168	48.441382	0%		0.041382	
48.5	2.361286386	-0.000061	0.014300%	0.007193	2.378513618	0.7296%	12689.5988	12688	0011000110010000	2.378927399	48.5471288	0%		0.047129	
48.6	2.370144865	-0.000061	0.014300%	0.007223	2.38750842	0.7326%	12737.587	12736	0011000111000000	2.387927124	48.6551255	0%		0.055125	
48.7	2.379712769	-0.000060	0.014400%	0.007254	2.397219781	0.7357%	12789.398	12788	0011000111110100	2.397676826	48.7721219	0%		0.072122	
48.8	2.388611651	-0.000059	0.014400%	0.007284	2.406255393	0.7387%	12837.6039	12836	0011001000100100	2.406676552	48.8801186	0%		0.080119	
48.9	2.398217515	-0.000059	0.014500%	0.007315	2.416007184	0.7418%	12889.6306	12891	0011001001011011	2.416988737	49.0038648	0%		0.103865	
49	2.407156358	-0.000058	0.014500%	0.007345	2.425084176	0.7448%	12938.0572	12941	0011001010001101	2.426363451	49.1163614	0%		0.116361	
49.1	2.416799739	-0.000058	0.014600%	0.007376	2.434874949	0.7479%	12990.292	12990	0011001010111110	2.435550671	49.226608	0%		0.126608	
49.2	2.425778099	-0.000057	0.014600%	0.007406	2.443992881	0.7509%	13038.937	13044	0011001011110100	2.445675362	49.3481043	0%		0.148104	
49.3	2.434776606	-0.000057	0.014700%	0.007437	2.453135141	0.7540%	13087.7118	13086	0011001100011110	2.453550121	49.4426015	0%		0.142601	
49.4	2.444475984	-0.000056	0.014700%	0.007467	2.462980628	0.7570%	13140.2385	13139	0011001101010011	2.463487318	49.5618478	0%		0.161848	
49.5	2.45351334	-0.000055	0.014800%	0.007498	2.472162177	0.7601%	13189.2229	13192	0011001110001000	2.473424515	49.6810942	0%		0.181094	
49.6	2.462570509	-0.000055	0.014800%	0.007528	2.481362771	0.7631%	13238.309	13238	0011001110110110	2.482049252	49.784591	0%		0.184591	
49.7	2.470969877	-0.000054	0.014900%	0.007559	2.489902613	0.7662%	13283.8698	13287	0011001111100111	2.491236471	49.8948377	0%		0.194838	
49.8	2.480743835	-0.000054	0.014900%	0.007589	2.499826578	0.7692%	13336.8151	13337	0011010000011001	2.500611185	50.0073342	0%		0.207334	
49.9	2.490535048	-0.000053	0.015000%	0.007620	2.509770054	0.7723%	13389.8645	13394	0011010001010010	2.511298359	50.1355803	0%		0.235580	
50	2.499669234	-0.000053	0.015000%	0.007650	2.519050526	0.7754%	13439.3768	13444	0011010010000100	2.520673073	50.2480769	0%		0.248077	
50.1	2.508822671	-0.000052	0.015100%	0.007681	2.528352501	0.7784%	13489.0037	13494	0011010010110110	2.530047787	50.3605734	1%		0.260573	
50.2	2.517995247	-0.000051	0.015100%	0.007711	2.537671786	0.7814%	13538.723	13543	0011010011100111	2.539235006	50.4708201	1%		0.270820	
50.3	2.527186848	-0.000051	0.015200%	0.007742	2.547014384	0.7846%	13588.5666	13589	0011010100010101	2.547859743	50.5743169	1%		0.274317	
50.4	2.536397362	-0.000050	0.015200%	0.007772	2.556373043	0.7876%	13638.496	13643	0011010101001011	2.557984434	50.6958132	1%		0.295813	
50.5	2.545626676	-0.000050	0.015300%	0.007803	2.565754808	0.7907%	13688.5486	13689	0011010101111001	2.566609171	50.7993101	1%		0.299310	
50.6	2.554874677	-0.000049	0.015300%	0.007833	2.575152391	0.7937%	13738.6856	13737	0011010110101001	2.575608896	50.9073068	1%		0.307307	
50.7	2.56414125	-0.000048	0.015400%	0.007864	2.584571867	0.7968%	13788.9394	13792	0011010111100000	2.585921082	51.031053	1%		0.331053	
50.8	2.573426283	-0.000048	0.015400%	0.007894	2.594008932	0.7998%	13839.2871	13838	0011011000001110	2.594545818	51.1345498	1%		0.334550	
50.9	2.582729662	-0.000047	0.015500%	0.007925	2.603466676	0.8029%	13889.745	13888	0011011001000000	2.603920532	51.2470464	1%		0.347046	
51	2.592051274	-0.000047	0.015500%	0.007955	2.612941764	0.8059%	13940.2955	13942	0011011001110110	2.614045223	51.3685427	1%		0.368543	
51.1	2.601391004	-0.000046	0.015600%	0.007986	2.622437325	0.8090%	13990.9553	13994	0011011010101010	2.623794926	51.4855391	1%		0.385539	
51.2	2.610087891	-0.000046	0.015600%	0.008016	2.631283797	0.8121%	14038.152	14037	0011011011010101	2.63185718	51.5822862	1%		0.382286	
51.3	2.618804888	-0.000045	0.015700%	0.008047	2.640152731	0.8152%	14085.4687	14087	0011011100000111	2.641231894	51.6947827	1%		0.394783	
51.4	2.628200508	-0.000044	0.015700%	0.008077	2.649703696	0.8182%	14136.424	14135	0011011100110111	2.650231619	51.8027794	1%		0.402779	
51.5	2.636956271	-0.000044	0.015800%	0.008108	2.658613842	0.8213%	14183.9605	14183	0011011101100111	2.659231344	51.9107761	1%		0.410776	
51.6	2.646388209	-0.000043	0.015800%	0.008138	2.668202569	0.8243%	14235.1172	14237	0011011111011101	2.669356035	52.0322724	1%		0.432272	
51.7	2.655837579	-0.000043	0.015900%	0.008169	2.677813161	0.8274%	14286.3907	14290	0011011111010010	2.679293232	52.1515188	1%		0.451519	
51.8	2.665304269	-0.000042	0.015900%	0.008199	2.687438098	0.8304%	14337.7406	14342	0011100000000110	2.689042934	52.2685152	1%		0.468515	
51.9	2.674788162	-0.000042	0.016000%	0.008230	2.697084692	0.8336%	14389.2061	14388	0011100000110100	2.697667671	52.3720121	1%		0.472012	
52	2.684289145	-0.000041	0.016000%	0.008260	2.706745383	0.8366%	14440.7469	14440	0011100001101000	2.707417374	52.4890085	1%		0.489008	
52.1	2.693156231	-0.000040	0.016100%	0.008291	2.715770201	0.8397%	14488.8951	14489	0011100010011001	2.716604593	52.5992551	1%		0.499255	
52.2	2.702692156	-0.000040	0.016100%	0.008321	2.725468295	0.8427%	14540.6354	14539	0011100011001011	2.725979307	52.7117517	1%		0.511752	
52.3	2.711596169	-0.000039	0.016200%	0.008352	2.734531491	0.8458%	14588.9884	14590	0011100011111110	2.735541515	52.8264982	1%		0.526498	
52.4	2.720519024	-0.000039	0.016200%	0.008382	2.743612498	0.8489%	14637.4365	14639	0011100100101111	2.744728735	52.9367448	1%		0.536745	
52.5	2.730107051	-0.000038	0.016300%	0.008413	2.753366636	0.8520%	14689.4757	14694	0011100101100110	2.75504092	53.060491	1%		0.560491	
52.6	2.73906614	-0.000038	0.016300%	0.008443	2.762485314	0.8550%	14738.1248	14741	0011100110010101	2.763853151	53.1662378	1%		0.566238	

52.7	2.748043725	-0.000037	0.016400%	0.008474	2.771624983	0.8581%	14786.8858	14789	0011100111000101	2.772852877	53.2742345	1%		0.574235	
52.8	2.757039689	-0.000036	0.016400%	0.008504	2.780780957	0.8611%	14835.7338	14840	0011100111111000	2.782415085	53.388981	1%		0.588981	
52.9	2.766053919	-0.000036	0.016500%	0.008535	2.78995973	0.8643%	14884.7034	14885	0011101000100101	2.790852327	53.4902279	1%		0.590228	
53	2.775086295	-0.000035	0.016500%	0.008565	2.799153548	0.8673%	14933.7533	14935	0011101001010111	2.800227041	53.6027245	1%		0.602724	
53.1	2.784136703	-0.000035	0.016600%	0.008596	2.808369952	0.8704%	14982.9237	14986	0011101010001010	2.809789249	53.717471	1%		0.617471	
53.2	2.793843728	-0.000034	0.016600%	0.008626	2.818245412	0.8734%	15035.6102	15035	0011101010111011	2.818976469	53.8277176	1%		0.627718	
53.3	2.802928744	-0.000034	0.016700%	0.008657	2.827497891	0.8766%	15084.9731	15086	0011101011101110	2.828538677	53.9424641	1%		0.642464	
53.4	2.812031443	-0.000033	0.016700%	0.008687	2.836764916	0.8796%	15134.4136	15133	0011101100011101	2.837350908	54.0482109	1%		0.648211	
53.5	2.8217871	-0.000032	0.016800%	0.008718	2.846694073	0.8827%	15187.3866	15191	0011101101010111	2.848225576	54.1787069	1%		0.678707	
53.6	2.83092371	-0.000032	0.016800%	0.008748	2.855997469	0.8857%	15237.0211	15238	0011101110000110	2.857037807	54.2844537	1%		0.684454	
53.7	2.840077654	-0.000031	0.016900%	0.008779	2.865320796	0.8888%	15286.7619	15285	0011101110110101	2.865850039	54.3902005	1%		0.690200	
53.8	2.849248813	-0.000031	0.016900%	0.008809	2.874660184	0.8919%	15336.5885	15336	0011101111101000	2.875412247	54.504947	1%		0.704947	
53.9	2.858437073	-0.000030	0.017000%	0.008840	2.884019291	0.8950%	15386.5202	15385	0011110000011001	2.884599466	54.6151936	1%		0.715194	
54	2.867642315	-0.000030	0.017000%	0.008870	2.893394207	0.8980%	15436.5363	15439	0011110001001111	2.894724157	54.7366899	1%		0.736690	
54.1	2.876864423	-0.000029	0.017100%	0.008901	2.902788628	0.9011%	15486.6564	15487	0011110001111111	2.903723883	54.8446866	1%		0.744687	
54.2	2.88610328	-0.000028	0.017100%	0.008931	2.912197599	0.9041%	15536.8542	15538	0011110010110010	2.913286091	54.9594331	1%		0.759433	
54.3	2.894105618	-0.000028	0.017200%	0.008962	2.920363395	0.9073%	15580.4195	15583	0011110011011111	2.921723333	55.06068	1%		0.760680	
54.4	2.903379823	-0.000027	0.017200%	0.008992	2.929809145	0.9103%	15630.8135	15629	0011110100001101	2.93034807	55.1641768	1%		0.764177	
54.5	2.912670425	-0.000027	0.017300%	0.009023	2.939275989	0.9134%	15681.32	15682	0011110101000010	2.940285267	55.2834232	1%		0.783423	
54.6	2.921977309	-0.000026	0.017300%	0.009053	2.94875587	0.9165%	15731.8961	15731	0011110101110011	2.949472486	55.3936698	1%		0.793670	
54.7	2.930678179	-0.000025	0.017400%	0.009084	2.957627742	0.9196%	15779.2284	15784	0011110110101000	2.959409683	55.5129162	1%		0.812916	
54.8	2.940018371	-0.000025	0.017400%	0.009114	2.967143387	0.9226%	15829.9953	15834	0011110111011010	2.968784397	55.6254128	2%		0.825413	
54.9	2.949374493	-0.000024	0.017500%	0.009145	2.97667768	0.9257%	15880.8616	15881	0011111000001001	2.977596628	55.7311595	2%		0.831160	
55	2.958746427	-0.000024	0.017500%	0.009175	2.986226523	0.9288%	15931.8056	15932	0011111000111100	2.987158836	55.845906	2%		0.845906	
55.1	2.968134057	-0.000023	0.017600%	0.009206	2.995793801	0.9319%	15982.848	15981	0011111001101101	2.996346056	55.9561527	2%		0.856153	
55.2	2.976920574	-0.000023	0.017600%	0.009236	3.004752937	0.9349%	16030.6458	16033	0011111010100001	3.006095758	56.0731491	2%		0.873149	
55.3	2.98634034	-0.000022	0.017700%	0.009267	3.01435382	0.9381%	16081.8675	16085	0011111011010101	3.015845461	56.1901455	2%		0.890146	
55.4	2.995160951	-0.000021	0.017700%	0.009297	3.023347473	0.9411%	16129.8495	16128	0011111100000000	3.023907715	56.2868926	2%		0.886893	
55.5	3.004612383	-0.000021	0.017800%	0.009328	3.032982503	0.9442%	16181.2533	16180	0011111100110100	3.033657417	56.403889	2%		0.903889	
55.6	3.013466616	-0.000020	0.017800%	0.009358	3.042011214	0.9472%	16229.4223	16234	0011111101101010	3.043782108	56.5253853	2%		0.925385	
55.7	3.022949244	-0.000020	0.017900%	0.009389	3.051678913	0.9504%	16281.0004	16283	0011111110011011	3.052969328	56.6356319	2%		0.935632	
55.8	3.031226505	-0.000019	0.017900%	0.009419	3.060126288	0.9534%	16326.068	16331	0011111111001011	3.061969053	56.7436286	2%		0.943629	
55.9	3.040131925	-0.000019	0.018000%	0.009450	3.069212435	0.9566%	16374.5434	16375	0011111111110111	3.070218801	56.8426256	2%		0.942626	
56	3.049054171	-0.000018	0.018000%	0.009480	3.078312019	0.9596%	16423.0906	16426	0100000000101010	3.07978101	56.9573721	2%		0.957372	
56.1	3.058599965	-0.000017	0.018100%	0.009511	3.088044808	0.9627%	16475.016	16480	0100000001100000	3.089905701	57.0788684	2%		0.978868	
56.2	3.067554415	-0.000017	0.018100%	0.009541	3.097179046	0.9657%	16523.748	16523	0100000010001011	3.097967955	57.1756155	2%		0.975615	
56.3	3.077129995	-0.000016	0.018200%	0.009572	3.106943099	0.9689%	16575.8402	16580	0100000011000100	3.108655128	57.3038615	2%		1.003862	
56.4	3.086116176	-0.000016	0.018200%	0.009602	3.116110512	0.9719%	16624.7492	16624	0100000011110000	3.116904877	57.4028585	2%		1.002859	
56.5	3.095118592	-0.000015	0.018300%	0.009633	3.125297019	0.9750%	16673.7601	16676	0100000100100100	3.126654579	57.5198549	2%		1.019855	
56.6	3.104137124	-0.000015	0.018300%	0.009663	3.134498219	0.9781%	16722.8494	16725	0100000101010101	3.135841799	57.6301016	2%		1.030102	
56.7	3.113171653	-0.000014	0.018400%	0.009694	3.143718296	0.9812%	16772.0394	16776	0100000110001000	3.145404007	57.7448481	2%		1.044848	
56.8	3.122222061	-0.000013	0.018400%	0.009724	3.1529518	0.9842%	16821.301	16825	0100000110111001	3.154591227	57.8550947	2%		1.055095	
56.9	3.13128823	-0.000013	0.018500%	0.009755	3.162205984	0.9874%	16870.673	16871	0100000111100111	3.163215963	57.9585916	2%		1.058592	
57	3.140370042	-0.000012	0.018500%	0.009785	3.171472331	0.9904%	16920.1098	16921	0100001000011001	3.172590677	58.0710881	2%		1.071088	
57.1	3.149467378	-0.000012	0.018600%	0.009816	3.18075914	0.9936%	16969.6558	16968	0100001001000100	3.181402908	58.1768349	2%		1.076835	
57.2	3.15858012	-0.000011	0.018600%	0.009846	3.190057859	0.9966%	17019.2654	17021	0100001001111101	3.191340105	58.2960813	2%		1.096081	
57.3	3.167114368	-0.000011	0.018700%	0.009877	3.198777117	0.9997%	17065.7835	17065	0100001010101001	3.199589853	58.3950782	2%		1.095078	
57.4	3.176258652	-0.000010	0.018700%	0.009907	3.208108811	1.0028%	17115.569	17119	0100001011011111	3.209714544	58.5165745	2%		1.116575	
57.5	3.184826374	-0.000009	0.018800%	0.009938	3.216861972	1.0059%	17162.2679	17167	0100001100001111	3.21871427	58.6245712	2%		1.124571	
57.6	3.194592256	-0.000009	0.018800%	0.009968	3.226823656	1.0089%	17215.4145	17220	0100001101000100	3.228651466	58.7438176	2%		1.143818	
57.7	3.203781339	-0.000008	0.018900%	0.009999	3.236205579	1.0121%	17265.4679	17266	0100001101110010	3.237276203	58.8473144	2%		1.147314	
57.8	3.212396753	-0.000008	0.018900%	0.010029	3.24500634	1.0151%	17312.4208	17316	0100001110100100	3.246650917	58.959811	2%		1.159811	
57.9	3.221616194	-0.000007	0.019000%	0.010060	3.254420077	1.0182%	17362.644	17366	0100001111010110	3.256025631	59.0723076	2%		1.172308	
58	3.230850096	-0.000007	0.019000%	0.010090	3.263846724	1.0213%	17412.9361	17417	0100010000001001	3.265587839	59.1870541	2%		1.187054	
58.1	3.239513216	-0.000006	0.019100%	0.010121	3.272699622	1.0244%	17460.1671	17463	0100010000110111	3.274212576	59.2905509	2%		1.190551	
58.2	3.248776766	-0.000005	0.019100%	0.010151	3.282156346	1.0275%	17510.6197	17510	0100010001100110	3.283024807	59.3962977	2%		1.196298	
58.3	3.257471456	-0.000005	0.019200%	0.010182	3.291043268	1.0306%	17558.0323	17559	0100010010010111	3.292212027	59.5065443	2%		1.206544	
58.4	3.266182291	-0.000004	0.019200%	0.010212	3.299942714	1.0336%	17605.5117	17609	0100010011001001	3.30158674	59.6190409	2%		1.219041	
58.5	3.275489964	-0.000004	0.019300%	0.010243	3.309450066	1.0368%	17656.2343	17658	0100010011111010	3.31077396	59.7292875	2%		1.229288	
58.6	3.284231654	-0.000003	0.019300%	0.010273	3.318381795	1.0398%	17703.8859	17707	0100010100101011	3.31996118	59.8395342	2%		1.239534	
58.7	3.294146453	-0.000002	0.019400%	0.010304	3.328502748	1.0429%	17757.8822	17756	0100010101011100	3.329148399	59.9497808	2%		1.249781	
58.8	3.302339864	-0.000002	0.019400%	0.010334	3.336882609	1.0460%	17802.5896	17802	0100010110001010	3.337773136	60.0532776	2%		1.253278	
58.9	3.311127485	-0.000001	0.019500%	0.010365	3.345865783	1.0491%	17850.5156	17853	0100010110111101	3.347335344	60.1680241	2%		1.268024	
59	3.319930537	-0.000001	0.019500%	0.010395	3.354862732	1.0522%	17898.5152	17897	0100010111101001	3.355585093	60.2670211	2%		1.267021	
59.1	3.329323263	0.000000	0.019600%	0.010426	3.364458503	1.0553%	17949.7096	17953	0100011000100001	3.366084772	60.3930173	2%		1.293017	
59.2	3.338155747	0.000000	0.019600%	0.010456	3.373486313	1.0584%	17997.8738	18002	0100011001010010	3.375271992	60.5032639	2%		1.303264	
59.3	3.347003304	0.000001	0.019700%	0.010487	3.382532312	1.0615%	18046.1351	18048	0100011010000000	3.383896729	60.6067607	2%		1.306761	
59.4	3.355865816	0.000002	0.019700%	0.010517	3.391590564	1.0645%	18094.4618	18095	0100011010101111	3.39270896	60.7125075	2%		1.312508	
59.5	3.364743165	0.000002	0.019800%	0.010548	3.400668808	1.0677%	18142.8951	18145	0100011011100001	3.402083673	60.8250041	2%		1.325004	
59.6	3.373635231	0.000003	0.019800%	0.010578	3.409758038	1.0707%	18191.387	18192	0100011100010000	3.410895905	60.9307509	2%		1.330751	
59.7	3.382541896	0.000003	0.019900%	0.010609	3.418867041	1.0739%	18239.9844	18245	0100011101000101	3.420833101	61.0499972	2%		1.349997	
59.8	3.392029917	0.000004	0.019900%	0.010639	3.428559747	1.0769%	18291.6959	18294	0100011101110110	3.430020321	61.1602439	2%		1.360244	
59.9	3.400964358	0.000004	0.020000%	0.010670	3.437697977	1.0801%	18340.4492	18342	0100011110100110	3.439020046	61.2682406	2%		1.368241	
60	3.4093483	0.000005	0.020000%	0.010700	3.446275831	1.0831%	18386.2129	18386	0100011111010010	3.447269794	61.3672375	2%		1.367238	
60.1	3.418312177	0.000006	0.020100%	0.010731	3.455443923	1.0863%	18435.1256	18435	0100100000000011	3.456457014	61.4774842	2%		1.377484	
60.2	3.427290061	0.000006	0.020100%	0.010761	3.464624258	1.0893%	18484.1035	18489	0100100000111001	3.466581705	61.5989805	2%		1.398980	
60.3	3.435720285	0.000007	0.020200%	0.010792	3.473254016	1.0925%	18530.1441	18535	0100100001100111	3.475206442	61.7024773	2%		1.402477	
60.4	3.444166406	0.000007	0.020200%	0.010822	3.481897877	1.0955%	18576.26	18577	0100100010010001	3.483081202	61.7969744	2%		1.396974	
60.5	3.453187727	0.000008	0.020300%	0.010853	3.491126291	1.0987%	18625.4944	18627	0100100011000011	3.492455915	61.909471	2%		1.409471	
60.6	3.462222582	0.000008	0.020300%	0.010883	3.500366439	1.1017%	18674.7915	18677	0100100011110101	3.501830629	62.0219676	2%		1.421968	
60.7	3.470713552	0.000009	0.020400%	0.010914	3.509059798	1.1049%	18721.1714	18722	0100100100100010	3.510267872	62.1232145	2%		1.423214	
60.8	3.480332426	0.000010	0.020400%	0.010944	3.518890529	1.1079%	18773.6193	18774	0100100101010110	3.520017574	62.2402109	2%		1.440211	

60.9	3.488851994	0.000010	0.020500%	0.010975	3.527614931	1.1111%	18820.1648	18825	0100100110001001	3.529579782	62.3549574	2%		1.454957	
61	3.497940869	0.000011	0.020500%	0.011005	3.536910935	1.1141%	18869.7599	18873	0100100110111001	3.538579508	62.4629541	2%		1.462954	
61.1	3.50648962	0.000011	0.020600%	0.011036	3.545665949	1.1173%	18916.4687	18917	0100100111100101	3.546829256	62.5619511	2%		1.461951	
61.2	3.515605328	0.000012	0.020600%	0.011066	3.554990208	1.1203%	18966.2146	18971	0100101000011011	3.556953947	62.6834474	2%		1.483447	
61.3	3.524182788	0.000012	0.020700%	0.011097	3.563775359	1.1235%	19013.0842	19013	0100101001000101	3.564828707	62.7779445	2%		1.477944	
61.4	3.532774952	0.000013	0.020700%	0.011127	3.572571307	1.1265%	19060.0114	19062	0100101001110110	3.574015926	62.8881911	2%		1.488191	
61.5	3.541930552	0.000014	0.020800%	0.011158	3.581941205	1.1296%	19110.0007	19112	0100101010101000	3.58339064	63.0006877	2%		1.500688	
61.6	3.550550714	0.000014	0.020800%	0.011188	3.590767573	1.1327%	19157.0902	19162	0100101011011010	3.592765354	63.1131842	2%		1.513184	
61.7	3.559731972	0.000015	0.020900%	0.011219	3.600164567	1.1358%	19207.2241	19206	0100101100000110	3.601015102	63.2121812	2%		1.512181	
61.8	3.567833961	0.000015	0.020900%	0.011249	3.608467965	1.1389%	19251.5235	19252	0100101100110100	3.609639839	63.3156781	2%		1.515678	
61.9	3.57649681	0.000016	0.021000%	0.011280	3.617341771	1.1420%	19298.8661	19297	0100101101100001	3.618077082	63.416925	2%		1.516925	
62	3.58517365	0.000017	0.021000%	0.011310	3.626226617	1.1451%	19346.2677	19346	0100101110010010	3.627264301	63.5271716	2%		1.527172	
62.1	3.594406913	0.000017	0.021100%	0.011341	3.635679496	1.1482%	19396.6997	19398	0100101111000110	3.637014004	63.644168	2%		1.544168	
62.2	3.603110333	0.000018	0.021100%	0.011371	3.644592333	1.1513%	19444.2505	19446	0100101111110110	3.646013729	63.7521647	2%		1.552165	
62.3	3.611286935	0.000018	0.021200%	0.011402	3.652977524	1.1545%	19488.9863	19490	0100110000100010	3.654263477	63.8511617	2%		1.551162	
62.4	3.620018559	0.000019	0.021200%	0.011432	3.661919965	1.1575%	19536.6951	19538	0100110001010010	3.663263203	63.9591584	2%		1.559158	
62.5	3.628763583	0.000019	0.021300%	0.011463	3.67088123	1.1607%	19584.5043	19588	0100110010000100	3.672637917	64.071655	3%		1.571655	
62.6	3.638059215	0.000020	0.021300%	0.011493	3.680395332	1.1637%	19635.263	19640	0100110010111000	3.682387619	64.1886514	3%		1.588651	
62.7	3.646829647	0.000021	0.021400%	0.011524	3.689382422	1.1668%	19683.2099	19686	0100110011100110	3.691012356	64.2921483	3%		1.592148	
62.8	3.655613128	0.000021	0.021400%	0.011554	3.698380547	1.1699%	19731.2158	19732	0100110100010100	3.699637093	64.3956451	3%		1.595645	
62.9	3.663875341	0.000022	0.021500%	0.011585	3.706854579	1.1731%	19776.4256	19777	0100110101000001	3.708074335	64.496892	3%		1.596892	
63	3.672685609	0.000022	0.021500%	0.011615	3.715880892	1.1761%	19824.5818	19827	0100110101110011	3.717449049	64.6093886	3%		1.609389	
63.1	3.681508574	0.000023	0.021600%	0.011646	3.724923356	1.1793%	19872.8242	19873	0100110110100001	3.726073786	64.7128854	3%		1.612885	
63.2	3.689281952	0.000023	0.021600%	0.011676	3.732901634	1.1823%	19915.3891	19914	0100110111000110	3.733761051	64.8051326	3%		1.605133	
63.3	3.698132034	0.000024	0.021700%	0.011707	3.741972629	1.1855%	19963.7838	19966	0100110111111110	3.743510754	64.922129	3%		1.622129	
63.4	3.707523476	0.000025	0.021700%	0.011737	3.751588186	1.1885%	20015.0837	20017	0100111000110001	3.753072962	65.0368755	3%		1.636876	
63.5	3.715869121	0.000025	0.021800%	0.011768	3.760150895	1.1917%	20060.7666	20061	0100111001011101	3.76132271	65.1358725	3%		1.635873	
63.6	3.724228949	0.000026	0.021800%	0.011798	3.768723667	1.1947%	20106.5031	20105	0100111010001001	3.769572458	65.2348695	3%		1.634869	
63.7	3.733128752	0.000026	0.021900%	0.011829	3.777848232	1.1979%	20155.1836	20159	0100111010111111	3.779697149	65.3563658	3%		1.656366	
63.8	3.741515555	0.000027	0.021900%	0.011859	3.78644937	1.2010%	20201.0714	20202	0100111011101010	3.787759403	65.4531128	3%		1.653113	
63.9	3.750440035	0.000027	0.022000%	0.011890	3.795600029	1.2041%	20249.8911	20251	0100111100011011	3.796946623	65.5633359	3%		1.663359	
64	3.758853343	0.000028	0.022000%	0.011920	3.804229063	1.2072%	20295.9278	20298	0100111101001010	3.805758854	65.6691062	3%		1.669106	
64.1	3.767280243	0.000029	0.022100%	0.011951	3.812876234	1.2103%	20342.0613	20342	0100111101110110	3.814008602	65.7681032	3%		1.668103	
64.2	3.776241381	0.000029	0.022100%	0.011981	3.822061808	1.2134%	20391.0672	20390	0100111110100110	3.823008327	65.8760999	3%		1.676100	
64.3	3.784694082	0.000030	0.022200%	0.012012	3.830736189	1.2165%	20437.3459	20438	0100111111010110	3.832008053	65.9840966	3%		1.684097	
64.4	3.793678736	0.000030	0.022200%	0.012042	3.839946675	1.2196%	20486.4847	20486	0101000000000110	3.841007778	66.0920933	3%		1.692093	
64.5	3.802156773	0.000031	0.022300%	0.012073	3.848647798	1.2228%	20532.906	20536	0101000000111000	3.850382492	66.2045899	3%		1.704590	
64.6	3.810647816	0.000031	0.022300%	0.012103	3.857359739	1.2258%	20579.3851	20583	0101000001100111	3.859194723	66.3103367	3%		1.710337	
64.7	3.818636104	0.000032	0.022400%	0.012134	3.86556618	1.2290%	20623.1672	20624	0101000010010000	3.866881989	66.4025839	3%		1.702584	
64.8	3.827153826	0.000033	0.022400%	0.012164	3.874305169	1.2320%	20669.7906	20671	0101000010111111	3.87569422	66.5083306	3%		1.708331	
64.9	3.835684202	0.000033	0.022500%	0.012195	3.883062441	1.2352%	20716.5115	20718	0101000011101110	3.884506451	66.6140774	3%		1.714077	
65	3.844227115	0.000034	0.022500%	0.012225	3.891828003	1.2382%	20763.2766	20764	0101000100011100	3.893131187	66.7175742	3%		1.717574	
65.1	3.852782448	0.000034	0.022600%	0.012256	3.900611633	1.2414%	20810.1381	20813	0101000101001101	3.902318407	66.8278209	3%		1.727821	
65.2	3.861350085	0.000035	0.022600%	0.012286	3.909403301	1.2445%	20857.0425	20862	0101000101111110	3.911505627	66.9380675	3%		1.738068	
65.3	3.86992991	0.000035	0.022700%	0.012317	3.918212823	1.2476%	20904.0421	20905	0101000110101001	3.919567881	67.0348146	3%		1.734815	
65.4	3.878013293	0.000036	0.022700%	0.012347	3.92651526	1.2507%	20948.3364	20947	0101000111010011	3.92744264	67.1293117	3%		1.729312	
65.5	3.886618161	0.000037	0.022800%	0.012378	3.935350208	1.2538%	20995.4717	20995	0101001000000011	3.936442366	67.2373084	3%		1.737308	
65.6	3.895234869	0.000037	0.022800%	0.012408	3.944194717	1.2569%	21042.658	21046	0101001000110110	3.946004574	67.3520549	3%		1.752055	
65.7	3.903863303	0.000038	0.022900%	0.012439	3.953054629	1.2601%	21089.9265	21091	0101001001100011	3.954441816	67.4533018	3%		1.753302	
65.8	3.912503348	0.000038	0.022900%	0.012469	3.961923853	1.2631%	21137.2447	21141	0101001010010101	3.96381653	67.5657984	3%		1.765798	
65.9	3.920651431	0.000039	0.023000%	0.012500	3.970298436	1.2663%	21181.9239	21182	0101001010111110	3.971503796	67.6580455	3%		1.758046	
66	3.92931536	0.000040	0.023000%	0.012530	3.979191908	1.2693%	21229.3714	21229	0101001011101101	3.980316027	67.7637923	3%		1.763792	
66.1	3.937489113	0.000040	0.023100%	0.012561	3.98759456	1.2725%	21274.2004	21276	0101001100011100	3.989128258	67.8695391	3%		1.769539	
66.2	3.945175597	0.000041	0.023100%	0.012591	3.995499197	1.2756%	21316.3724	21317	0101001101000101	3.996815523	67.9617863	3%		1.761786	
66.3	3.953875995	0.000041	0.023200%	0.012622	4.004436257	1.2788%	21364.0524	21367	0101001101110111	4.006190237	68.0742828	3%		1.774283	
66.4	3.962587315	0.000042	0.023200%	0.012652	4.013379863	1.2818%	21411.7674	21412	0101001110100100	4.01462748	68.1755298	3%		1.775530	
66.5	3.970812019	0.000042	0.023300%	0.012683	4.021836232	1.2850%	21456.883	21461	0101001111010101	4.023814699	68.2857764	3%		1.785776	
66.6	3.979049419	0.000043	0.023300%	0.012713	4.030300897	1.2880%	21502.0428	21503	0101001111111111	4.031689459	68.3802735	3%		1.780274	
66.7	3.987794823	0.000044	0.023400%	0.012744	4.039284728	1.2912%	21549.9724	21552	0101010000110000	4.040876678	68.4905201	3%		1.790520	
66.8	3.99605627	0.000044	0.023400%	0.012774	4.047775817	1.2943%	21595.2732	21596	0101010001011100	4.049126427	68.5895171	3%		1.789517	
66.9	4.004330067	0.000045	0.023500%	0.012805	4.056283028	1.2974%	21640.66	21643	0101010001001011	4.057938658	68.6952639	3%		1.795264	
67	4.013108526	0.000045	0.023500%	0.012835	4.065298893	1.3005%	21688.7605	21691	0101010010111011	4.066938383	68.8032606	3%		1.803261	
67.1	4.02140568	0.000046	0.023600%	0.012866	4.073830841	1.3037%	21734.2792	21735	0101010011100111	4.075188131	68.9022576	3%		1.802258	
67.2	4.029714839	0.000046	0.023600%	0.012896	4.082372365	1.3067%	21779.8491	21782	0101010100010110	4.084000362	69.0080043	3%		1.808004	
67.3	4.038035887	0.000047	0.023700%	0.012927	4.090929584	1.3099%	21825.5027	21826	0101010101000010	4.092250111	69.1070013	3%		1.807001	
67.4	4.045880258	0.000048	0.023700%	0.012957	4.099000253	1.3129%	21868.5605	21872	0101010101110000	4.100874847	69.2104982	3%		1.810498	
67.5	4.054225732	0.000048	0.023800%	0.012988	4.107584283	1.3161%	21914.3571	21916	0101010110011100	4.109124596	69.3094951	3%		1.809495	
67.6	4.062582751	0.000049	0.023800%	0.013018	4.116175366	1.3192%	21960.1913	21964	0101010111001100	4.118124321	69.4174919	3%		1.817492	
67.7	4.071436682	0.000049	0.023900%	0.013049	4.125275647	1.3224%	22008.7422	22007	0101010111110111	4.126186575	69.5142389	3%		1.814239	
67.8	4.078846472	0.000050	0.023900%	0.013079	4.132908017	1.3254%	22049.4616	22049	0101011000100001	4.134061335	69.608736	3%		1.808736	
67.9	4.087238427	0.000050	0.024000%	0.013110	4.141541272	1.3286%	22095.5209	22094	0101011001001110	4.142498577	69.7099829	3%		1.809983	
68	4.095158949	0.000051	0.024000%	0.013140	4.149692142	1.3317%	22139.0066	22138	0101011001111010	4.150748325	69.8089799	3%		1.808980	
68.1	4.103573951	0.000052	0.024100%	0.013171	4.158348799	1.3348%	22185.1907	22188	0101011010101100	4.160123039	69.9214765	3%		1.821476	
68.2	4.11151926	0.000052	0.024100%	0.013201	4.166526815	1.3379%	22228.8212	22229	0101011011010101	4.167810305	70.0137237	3%		1.813724	
68.3	4.120436425	0.000053	0.024200%	0.013232	4.175693433	1.3410%	22277.7259	22276	0101011100000100	4.176622536	70.1194704	3%		1.819470	
68.4	4.12840508	0.000053	0.024200%	0.013262	4.183896141	1.3441%	22321.4882	22325	0101011100110101	4.185809755	70.2297171	3%		1.829717	
68.5	4.13686383	0.000054	0.024300%	0.013293	4.192599281	1.3473%	22367.9203	22367	0101011101011111	4.193684515	70.3242142	3%		1.824214	
68.6	4.144856355	0.000054	0.024300%	0.013323	4.200827203	1.3504%	22411.817	22412	0101011110001100	4.202121758	70.4254611	3%		1.825461	
68.7	4.152861133	0.000055	0.024400%	0.013354	4.209071296	1.3535%	22455.8001	22457	0101011110111001	4.210559	70.526708	3%		1.826708	
68.8	4.160878048	0.000056	0.024400%	0.013384	4.217323936	1.3566%	22499.8287	22504	0101011111101000	4.219371231	70.6324548	3%		1.832455	
68.9	4.169380693	0.000056	0.024500%	0.013415	4.226074708	1.3598%	22546.5149	22547	0101100000010011	4.227433485	70.7292018	3%		1.829202	
69	4.177893301	0.000057	0.024500%	0.013445	4.234830822	1.3628%	22593.2296	22595	0101100001000011	4.23643321	70.8371985	3%		1.837199	

69.1	4.185943996	0.000057	0.024600%	0.013476	4.243124561	1.3660%	22637.4775	22636	0101100001101100	4.244120476	70.9294457	3%		1.829446	
69.2	4.194006373	0.000058	0.024600%	0.013506	4.25142534	1.3691%	22681.763	22685	0101100010011101	4.253307695	71.0396923	3%		1.839692	
69.3	4.201610495	0.000058	0.024700%	0.013537	4.259267413	1.3723%	22723.6012	22724	0101100011000100	4.260619972	71.1274397	3%		1.827440	
69.4	4.209696863	0.000059	0.024700%	0.013567	4.267593517	1.3753%	22768.0217	22768	0101100011110000	4.26886972	71.2264366	3%		1.826437	
69.5	4.21779457	0.000060	0.024800%	0.013598	4.275935959	1.3785%	22812.5295	22817	0101100100100001	4.27805694	71.3366833	3%		1.836683	
69.6	4.225903504	0.000060	0.024800%	0.013628	4.284286976	1.3816%	22857.0829	22856	0101100101001000	4.285369217	71.4244306	3%		1.824431	
69.7	4.234023552	0.000061	0.024900%	0.013659	4.292653108	1.3847%	22901.7171	22903	0101100101110111	4.294181448	71.5301774	3%		1.830177	
69.8	4.2421546	0.000061	0.024900%	0.013689	4.301027573	1.3878%	22946.3956	22949	0101100110100101	4.302806185	71.6336742	3%		1.833674	
69.9	4.249832509	0.000062	0.025000%	0.013720	4.308946458	1.3910%	22988.6436	22987	0101100111001011	4.309930967	71.7191716	3%		1.819172	
70	4.258449253	0.000063	0.025000%	0.013750	4.317813389	1.3940%	23035.9496	23035	0101100111111011	4.318930693	71.8271683	3%		1.827168	
70.1	4.266612633	0.000063	0.025100%	0.013781	4.326226562	1.3972%	23080.8347	23085	0101101000101101	4.328305407	71.9396649	3%		1.839665	
70.2	4.274325413	0.000064	0.025100%	0.013811	4.334177944	1.4003%	23123.2561	23125	0101101001010101	4.335805178	72.0296621	3%		1.829662	
70.3	4.282510746	0.000064	0.025200%	0.013842	4.342614412	1.4035%	23168.2654	23169	0101101010000001	4.344054926	72.1286591	3%		1.828659	
70.4	4.290247169	0.000065	0.025200%	0.013872	4.350590759	1.4065%	23210.82	23215	0101101010101111	4.352679663	72.232156	3%		1.832156	
70.5	4.298454004	0.000065	0.025300%	0.013903	4.359050071	1.4097%	23255.9512	23256	0101101011011000	4.360366928	72.3244031	3%		1.824403	
70.6	4.306670948	0.000066	0.025300%	0.013933	4.367514722	1.4128%	23301.111	23301	0101101100000101	4.368804171	72.42565	3%		1.825650	
70.7	4.314441509	0.000067	0.025400%	0.013964	4.375531622	1.4159%	23343.8819	23347	0101101100110011	4.377428907	72.5291469	3%		1.829147	
70.8	4.321768431	0.000067	0.025400%	0.013994	4.38309566	1.4190%	23384.2368	23388	0101101101011100	4.385116173	72.6213941	3%		1.821394	
70.9	4.330017884	0.000068	0.025500%	0.014025	4.39159925	1.4222%	23429.6042	23431	0101101110000111	4.393178427	72.7181411	3%		1.818141	
71	4.338277002	0.000068	0.025500%	0.014055	4.400109729	1.4253%	23475.0085	23478	0101101110110110	4.401990658	72.8238879	3%		1.823888	
71.1	4.346093091	0.000069	0.025600%	0.014086	4.408174824	1.4284%	23518.0365	23520	0101101111100000	4.409865417	72.918385	3%		1.818385	
71.2	4.353920518	0.000069	0.025600%	0.014116	4.416248462	1.4315%	23561.1102	23562	0101110000001010	4.417740177	73.0128821	3%		1.812882	
71.3	4.36175917	0.000070	0.025700%	0.014147	4.424337451	1.4347%	23604.2657	23609	0101110000111001	4.426552408	73.1186289	3%		1.818629	
71.4	4.369608935	0.000071	0.025700%	0.014177	4.432433726	1.4378%	23647.4601	23648	0101110001100000	4.433864685	73.2063762	3%		1.806376	
71.5	4.377918511	0.000071	0.025800%	0.014208	4.44100245	1.4410%	23693.1751	23698	0101110010010010	4.443239399	73.3188728	3%		1.818873	
71.6	4.385789227	0.000072	0.025800%	0.014238	4.449120986	1.4440%	23736.4882	23738	0101110010111010	4.45073917	73.40887	3%		1.808870	
71.7	4.393223792	0.000072	0.025900%	0.014269	4.456803087	1.4472%	23777.473	23779	0101110011100011	4.458426435	73.5011172	3%		1.801117	
71.8	4.401116893	0.000073	0.025900%	0.014299	4.464945317	1.4503%	23820.9126	23822	0101110100001110	4.466488689	73.5978643	3%		1.797864	
71.9	4.409020552	0.000073	0.026000%	0.014330	4.473104305	1.4535%	23864.4415	23869	0101110100111101	4.47530092	73.703611	3%		1.803611	
72	4.417378778	0.000074	0.026000%	0.014360	4.481719436	1.4565%	23910.4041	23910	0101110101100110	4.482988186	73.7958582	2%		1.795858	
72.1	4.424859096	0.000075	0.026100%	0.014391	4.489448921	1.4597%	23951.6416	23950	0101110110000110	4.490487957	73.8858555	2%		1.785855	
72.2	4.432793763	0.000075	0.026100%	0.014421	4.49763634	1.4628%	23995.3223	23996	0101110110111100	4.499112694	73.9893523	2%		1.789352	
72.3	4.440297222	0.000076	0.026200%	0.014452	4.505390284	1.4660%	24036.6903	24038	0101110111100110	4.506987453	74.0838494	2%		1.783849	
72.4	4.448252942	0.000076	0.026200%	0.014482	4.513600058	1.4691%	24080.4903	24082	0101111000010010	4.515237202	74.1828464	2%		1.782846	
72.5	4.455779093	0.000077	0.026300%	0.014513	4.52137801	1.4722%	24121.9864	24121	0101111000111001	4.522549479	74.2705937	2%		1.770594	
72.6	4.463755425	0.000077	0.026300%	0.014543	4.529609696	1.4753%	24165.9032	24169	0101111001101001	4.531549204	74.3785904	2%		1.778590	
72.7	4.471303823	0.000078	0.026400%	0.014574	4.537411209	1.4785%	24207.5251	24212	0101111010010100	4.539611458	74.4753375	2%		1.775337	
72.8	4.478863641	0.000079	0.026400%	0.014604	4.545220189	1.4815%	24249.1867	24254	0101111010111110	4.547486217	74.5698346	2%		1.769835	
72.9	4.48687053	0.000079	0.026500%	0.014635	4.553488992	1.4847%	24293.3016	24292	0101111011100100	4.554611	74.655332	2%		1.755332	
73	4.494886771	0.000080	0.026500%	0.014665	4.561762162	1.4878%	24337.4397	24340	0101111100010100	4.563610725	74.7633287	2%		1.763329	
73.1	4.502478335	0.000080	0.026600%	0.014696	4.569610478	1.4910%	24379.3113	24383	0101111100111111	4.571672979	74.8600758	2%		1.760076	
73.2	4.510080875	0.000081	0.026600%	0.014726	4.577464767	1.4941%	24421.2146	24425	0101111101101001	4.579547739	74.9545729	2%		1.754573	
73.3	4.518126366	0.000081	0.026700%	0.014757	4.585774789	1.4973%	24465.5494	24467	0101111110010011	4.587422499	75.04907	2%		1.749070	
73.4	4.52574961	0.000082	0.026700%	0.014787	4.593651071	1.5003%	24507.5701	24509	0101111110111101	4.595297258	75.1435671	2%		1.743567	
73.5	4.533383501	0.000083	0.026800%	0.014818	4.60154331	1.5035%	24549.676	24553	0101111111101001	4.603547007	75.2425641	2%		1.742564	
73.6	4.541027928	0.000083	0.026800%	0.014848	4.609443077	1.5066%	24591.822	24594	0110000000010010	4.611234272	75.3348113	2%		1.734811	
73.7	4.548254354	0.000084	0.026900%	0.014879	4.616922684	1.5098%	24631.7264	24635	0110000000111011	4.618921537	75.4270584	2%		1.727058	
73.8	4.556347954	0.000084	0.026900%	0.014909	4.62527938	1.5129%	24676.3102	24679	0110000001100111	4.627171286	75.5260554	2%		1.726055	
73.9	4.564023335	0.000085	0.027000%	0.014940	4.633215711	1.5160%	24718.6513	24719	0110000010001111	4.634671057	75.6160527	2%		1.716053	
74	4.570857421	0.000086	0.027000%	0.014970	4.64029375	1.5191%	24756.4133	24760	0110000010111000	4.642358322	75.7082999	2%		1.708300	
74.1	4.578554707	0.000086	0.027100%	0.015001	4.6482543	1.5223%	24798.8836	24801	0110000011100001	4.650045588	75.8005471	2%		1.700547	
74.2	4.586685761	0.000087	0.027100%	0.015031	4.656649996	1.5254%	24843.6755	24847	0110000100001111	4.658670324	75.9040439	2%		1.704044	
74.3	4.593978818	0.000087	0.027200%	0.015062	4.664201142	1.5286%	24883.9615	24886	0110000100110110	4.665982601	75.9917912	2%		1.691791	
74.4	4.601705426	0.000088	0.027200%	0.015092	4.6721872	1.5316%	24926.5679	24925	0110000101011101	4.673294878	76.0795385	2%		1.679539	
74.5	4.609020417	0.000088	0.027300%	0.015123	4.679761574	1.5348%	24966.9779	24969	0110000110001001	4.681544626	76.1785355	2%		1.678536	
74.6	4.616766937	0.000089	0.027300%	0.015153	4.687768819	1.5379%	25009.6974	25008	0110000110110000	4.688856903	76.2662828	2%		1.666283	
74.7	4.624522802	0.000090	0.027400%	0.015184	4.695790809	1.5411%	25052.4955	25051	0110000111011011	4.696919157	76.3630299	2%		1.663030	
74.8	4.631869424	0.000090	0.027400%	0.015214	4.703393978	1.5442%	25093.0591	25095	0110001000000111	4.705168905	76.4620269	2%		1.662027	
74.9	4.639226971	0.000091	0.027500%	0.015245	4.711012472	1.5474%	25133.7045	25138	0110001000110010	4.713231159	76.5587739	2%		1.658774	
75	4.647012023	0.000091	0.027500%	0.015275	4.719061806	1.5505%	25176.6485	25178	0110001001011010	4.72073093	76.6487712	2%		1.648771	
75.1	4.654390197	0.000092	0.027600%	0.015306	4.726702214	1.5536%	25217.4108	25219	0110001010000011	4.728418196	76.7410183	2%		1.641018	
75.2	4.66177897	0.000092	0.027600%	0.015336	4.734350073	1.5567%	25258.2128	25262	0110001010101110	4.73648045	76.8377654	2%		1.637765	
75.3	4.669178231	0.000093	0.027700%	0.015367	4.742012851	1.5599%	25299.0945	25298	0110001011010010	4.743230244	76.9187629	2%		1.618763	
75.4	4.677000998	0.000094	0.027700%	0.015397	4.750101417	1.5630%	25342.2478	25345	0110001100000001	4.752042475	77.0245097	2%		1.624510	
75.5	4.684420025	0.000094	0.027800%	0.015428	4.757786256	1.5662%	25383.2471	25384	0110001100101000	4.759354752	77.112257	2%		1.612257	
75.6	4.691849218	0.000095	0.027800%	0.015458	4.76547605	1.5692%	25424.2729	25424	0110001101010000	4.766854523	77.2022543	2%		1.602254	
75.7	4.698878004	0.000095	0.027900%	0.015489	4.77276547	1.5724%	25463.1627	25462	0110001101110110	4.773979305	77.2877517	2%		1.587752	
75.8	4.706328091	0.000096	0.027900%	0.015519	4.78047742	1.5755%	25504.3067	25507	0110001110100011	4.782416548	77.3889986	2%		1.588999	
75.9	4.713788021	0.000096	0.028000%	0.015550	4.788205718	1.5787%	25545.5379	25548	0110001111001100	4.790103813	77.4812458	2%		1.581246	
76	4.720849869	0.000097	0.028000%	0.015580	4.795524233	1.5818%	25584.5829	25586	0110001111110010	4.797228596	77.5667431	2%		1.566743	
76.1	4.728736985	0.000098	0.028100%	0.015611	4.803686428	1.5850%	25628.129	25632	0110010000100000	4.805853333	77.67024	2%		1.570240	
76.2	4.736225807	0.000098	0.028100%	0.015641	4.811440643	1.5881%	25669.4984	25674	0110010001001010	4.813728092	77.7647371	2%		1.564737	
76.3	4.743318862	0.000099	0.028200%	0.015672	4.818797142	1.5913%	25708.7461	25713	0110010001110001	4.821040369	77.8524844	2%		1.552484	
76.4	4.750422983	0.000099	0.028200%	0.015702	4.826161459	1.5944%	25748.0354	25749	0110010010010101	4.827790163	77.933482	2%		1.533482	
76.5	4.75834493	0.000100	0.028300%	0.015733	4.834361019	1.5975%	25791.7808	25790	0110010010111110	4.835477428	78.0257291	2%		1.525729	
76.6	4.765469106	0.000100	0.028300%	0.015763	4.841746644	1.6006%	25831.1839	25835	0110010011101011	4.843914671	78.1269761	2%		1.526976	
76.7	4.772604022	0.000101	0.028400%	0.015794	4.849147542	1.6038%	25870.6684	25871	0110010100001111	4.850664465	78.2079736	2%		1.507974	
76.8	4.780150394	0.000102	0.028400%	0.015824	4.856962044	1.6069%	25912.3595	25915	0110010100111011	4.858914213	78.3069706	2%		1.506971	
76.9	4.787305599	0.000102	0.028500%	0.015855	4.864385512	1.6101%	25951.9644	25954	0110010101100010	4.86622649	78.3947179	2%		1.494718	
77	4.794471223	0.000103	0.028500%	0.015885	4.871814068	1.6132%	25991.5965	25994	0110010110001010	4.873726261	78.4847151	2%		1.484715	
77.1	4.801647157	0.000103	0.028600%	0.015916	4.879259529	1.6164%	26031.3187	26035	0110010110110011	4.881413527	78.5769623	2%		1.476962	
77.2	4.809230655	0.000104	0.028600%	0.015946	4.887113644	1.6194%	26073.2212	26073	0110010111011001	4.888538309	78.6624597	2%		1.462460	

	77.3	4.816426028	0.000104	0.028700%	0.015977	4.894579803	1.6227%	26113.0539	26112	0110011000000000	4.895850586	78.750207	2%		1.450207	
	77.4	4.82323576	0.000105	0.028700%	0.016007	4.901648519	1.6257%	26150.7661	26152	0110011000101000	4.903350357	78.8402043	2%		1.440204	
	77.5	4.830846648	0.000106	0.028800%	0.016038	4.90953687	1.6289%	26192.8512	26195	0110011001010011	4.911412611	78.9369513	2%		1.436951	
	77.6	4.838071682	0.000106	0.028800%	0.016068	4.917029581	1.6320%	26232.8256	26234	0110011001111010	4.918724888	79.0246987	2%		1.424699	
	77.7	4.844913332	0.000107	0.028900%	0.016099	4.924137081	1.6352%	26270.7448	26269	0110011010011101	4.925287188	79.1034463	2%		1.403446	
	77.8	4.852158468	0.000107	0.028900%	0.016129	4.931651134	1.6383%	26310.833	26312	0110011011001000	4.933349442	79.2001933	2%		1.400193	
	77.9	4.85941307	0.000108	0.029000%	0.016160	4.93917927	1.6415%	26350.9963	26350	0110011011101110	4.940474224	79.2856907	2%		1.385691	
	78	4.866677032	0.000109	0.029000%	0.016190	4.946712366	1.6446%	26391.1861	26393	0110011100011001	4.948536478	79.3824377	2%		1.382438	
	78.1	4.873560604	0.000109	0.029100%	0.016221	4.953865313	1.6478%	26429.3477	26428	0110011100111100	4.955098778	79.4611853	2%		1.361185	
	78.2	4.881232623	0.000110	0.029100%	0.016251	4.961814122	1.6508%	26471.7554	26473	0110011101101001	4.96353602	79.5624322	2%		1.362432	
	78.3	4.888136096	0.000110	0.029200%	0.016282	4.968988218	1.6540%	26510.0299	26515	0110011110010011	4.97141078	79.6569294	2%		1.356929	
	78.4	4.89505021	0.000111	0.029200%	0.016312	4.976167508	1.6571%	26548.3321	26552	0110011110111000	4.978348068	79.7401768	2%		1.340177	
	78.5	4.902361114	0.000111	0.029300%	0.016343	4.983756701	1.6603%	26588.8212	26592	0110011111100000	4.985847839	79.8301741	2%		1.330174	
	78.6	4.909680757	0.000112	0.029300%	0.016373	4.991349167	1.6634%	26629.3277	26632	0110100000001000	4.99334761	79.9201713	2%		1.320171	
	78.7	4.916624467	0.000113	0.029400%	0.016404	4.998564975	1.6666%	26667.8247	26671	0110100000101111	5.000659887	80.0079186	2%		1.307919	
	78.8	4.923962124	0.000113	0.029400%	0.016434	5.006177695	1.6697%	26708.4393	26707	0110100001010011	5.007409681	80.0889162	2%		1.288916	
	78.9	4.930925325	0.000114	0.029500%	0.016465	5.013414236	1.6729%	26747.047	26749	0110100001111101	5.015284441	80.1834133	2%		1.283413	
	79	4.937898535	0.000114	0.029500%	0.016495	5.020657321	1.6760%	26785.6895	26789	0110100010100101	5.022784212	80.2734105	2%		1.273411	
	79.1	4.944500436	0.000115	0.029600%	0.016526	5.027527395	1.6792%	26822.342	26822	0110100011000110	5.028971523	80.3476583	2%		1.247658	
	79.2	4.951494184	0.000115	0.029600%	0.016556	5.034792245	1.6823%	26861.1007	26865	0110100011110001	5.037033777	80.4444053	2%		1.244405	
	79.3	4.958497626	0.000116	0.029700%	0.016587	5.042071504	1.6855%	26899.9363	26900	0110100100010100	5.043596077	80.5231529	2%		1.223153	
	79.4	4.965889368	0.000117	0.029700%	0.016617	5.049740949	1.6886%	26940.8535	26940	0110100100111100	5.051095848	80.6131502	2%		1.213150	
	79.5	4.972533175	0.000117	0.029800%	0.016648	5.056656426	1.6918%	26977.7482	26977	0110100101100001	5.058033136	80.6963976	2%		1.196398	
	79.6	4.979565076	0.000118	0.029800%	0.016678	5.063960847	1.6948%	27016.718	27016	0110100110001000	5.065345413	80.784145	1%		1.184145	
	79.7	4.986606256	0.000118	0.029900%	0.016709	5.071281327	1.6981%	27055.7735	27059	0110100110110011	5.073407667	80.880892	1%		1.180892	
	79.8	4.993656614	0.000119	0.029900%	0.016739	5.078605419	1.7011%	27094.8482	27097	0110100111011001	5.08053245	80.9663894	1%		1.166389	
	79.9	5.000341479	0.000119	0.030000%	0.016770	5.085564422	1.7043%	27131.9752	27137	0110101000000001	5.088032221	81.0563867	1%		1.156387	
	80	5.002267689	0.000120	0.030000%	0.016800	5.0876777	1.7074%	27143.2497	27142	0110101000000110	5.088969692	81.0676363	1%		1.067636	